

ActewAGL Distribution 2013/14 Network Pricing Proposal

Submission to the Australian Energy Regulator

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Overview

ActewAGL Distribution offers customers a range of network tariff options across three tariff classes—residential, commercial low voltage and high voltage. Customers are able to choose the tariff that best suits their needs, subject to some eligibility requirements as set out in this document.

Tariff structures and customer choices in the National Electricity Market (NEM) have been the subject of intense scrutiny over the past year, through several major reviews including the Australian Energy Market Commission's (AEMC's) Power of Choice review, the Productivity Commission's Electricity Networks Regulatory Frameworks review and the review by the Senate Select Committee on Electricity Prices. ActewAGL Distribution currently offers the types of tariffs that have been recommended in these reviews, including time-of-use (TOU) tariffs, as the default for all new customers, and demand and capacity charges for commercial customers.

ActewAGL Distribution proposes to change the buy-back arrangements for consumers installing small embedded PV generation systems (less than 30KW) and not receiving the ACT feed-in tariff. Throughout 2013/14 ActewAGL Distribution will continue to review its tariff structure to ensure it meets the diverse needs of customers and complies with any changes to the National Electricity Rules arising from recommendations by the AEMC, the Productivity Commission and others.

Electricity prices, and the role of network charges as a driver of recent increases, have been the subject of ongoing debate. ActewAGL Distribution's network charges remain the lowest in the NEM, according to AEMC estimates.¹ Significant investment in the network over the 2009-14 regulatory period means that network prices must increase to generate the required revenue, as the AER explains in the *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision* (the Final Distribution Determination). However, the AER approved increase in revenue for ActewAGL Distribution, 34 per cent (in real terms), is below the 44 per cent approved increase across the NEM.²

ActewAGL Distribution estimates that the proposed 2013/14 network and metering charges will raise the electricity bill for an average residential customer, consuming 7000 kWh on the residential basic network charge, by \$0.71 per week (including GST)—a real increase of 3.9 per cent (5.7 per cent nominal). For a commercial customer consuming 30 MWh per annum on the general network charge, the proposed network and metering price increases would raise the electricity bill by \$3.69 per week (including GST)—a 3.4 per cent real increase (5.2 per cent nominal).

¹ AEMC 2013, *Electricity Price Trends Final Report*, March, chapter 3

² AER 2012, *State of the Energy Market 2012*, December, pp. 9-10

The proposed distribution use of system (DUOS) charges for 2013/14 are 0.54 cents per kWh, or 9.2 per cent in nominal terms, higher on average than the charges for 2012/13. TUOS charges, levied on ActewAGL Distribution by TransGrid, are 0.24 cents per kWh, or 14.9 per cent in nominal terms, lower on average than the charges for 2012/13. The proposed network use of system (NUOS) charges (comprising both DUOS and TUOS) for 2013/14 are, as a result, on average 0.30 c/kWh, or 4.0 per cent in nominal terms, higher than the average charges for 2012/13. Charges for miscellaneous services for 2013/14 average about 3.9 per cent higher than the 2012/13 charges.

Charges for metering services are subject to a separate regulatory cap. Most of the proposed metering charges for 2013/14 are about 1.7 per cent higher (in nominal terms) than the charges for 2012/13.

1 Introduction

1.1 Purpose and scope of the document

ActewAGL Distribution has prepared this document in accordance with the requirements in Chapter 11 of the *National Electricity Amendment Rules* (the transitional *Rules*). It provides the required information on the tariffs and charges to apply to ActewAGL Distribution's regulated distribution services from 1 July 2013 to 30 June 2014. A checklist of the transitional *Rules* requirements and where they are met in this document is provided in Attachment 1.

The document relates to tariffs and charges for ActewAGL Distribution's standard control services and alternative control services, as defined in clause 6.2.3C of the transitional *Rules*. ActewAGL Distribution's standard control services comprise distribution network use of system services and miscellaneous distribution services. ActewAGL Distribution's alternative control services comprise the provision and servicing of all manually read meters for consumers with loads less than 160 MWh per annum.

In accordance with the transitional *Rules*, separate regulatory control mechanisms apply to standard control and alternative control services. Therefore, separate price schedules must be determined. The combined standard control (network) and alternative control (metering) prices are also provided in this document.

The retail component of the delivered price of electricity to consumers accounts for around 60 per cent of the total price. The retail provision of electricity in the ACT is fully open to competition, although there remains a regulated retail tariff offer, set through a separate retail regulatory process, for customers consuming less than 100 MWh per year.

As well as setting out the proposed network tariffs and charges and demonstrating compliance with the relevant transitional *Rules* and the 2009-14 Final Distribution Determination, the pricing proposal includes explanations of the basis for the current tariff structure and the tariff setting process. While this information is not required under the pricing provisions in the *Rules*, ActewAGL Distribution believes that transparency and promoting consumer awareness are important, and the annual network pricing proposal provides a useful vehicle for achieving this. The recent reviews by the AEMC (*Power of Choice*) and the Productivity Commission (*Electricity Networks Regulatory Frameworks*), both of which ActewAGL Distribution participated in, highlighted the importance of customer engagement in relation to network tariffs.

1.2 Background

The Australian Energy Regulator (the AER) is responsible for the economic regulation of distribution services provided by ActewAGL Distribution. The AER has determined the maximum allowed average revenue (MAAR) for ActewAGL Distribution's standard control services and a revenue cap for alternative control (metering) services for the 2009-14

regulatory period.³ In May 2012 the AER accepted ActewAGL Distribution's request to revoke and substitute the determination, in accordance with clause 6.13 of the transitional Rules. The X factor for standard control services for 2013/14 (and 2012/13) has changed from -4.00 per cent to -5.41 per cent.

This document should be read in conjunction with the AER's Final Distribution Determination⁴ and ActewAGL Distribution's submissions and proposals (all published on the AER's website), as they set out in detail the basis of the costs that are reflected in the prices in this document. The AER summarises the outcomes of the regulatory review as follows:

Over the course of the next regulatory control period, ActewAGL will significantly increase investment on its network, which will result in higher prices for electricity consumers in the ACT. Higher prices will be largely driven by significant investment in four major capital projects. These projects include construction of two new zone substations, which are the first to be built in the ACT since 1994, augmentation of a third substation and construction of new assets to improve the security of electricity supply to the ACT. In addition, ActewAGL has already undertaken significant capital works to reinforce and replace a large number of unsafe poles, and these costs will now flow through to consumers. While consumers within the ACT will face higher charges as a result of this investment, they will also benefit from a more reliable and secure network.⁵

The AER also notes that, in addition to higher capital investment, increases in network charges are being driven by the introduction of the feed-in tariff scheme, which imposes costs on ActewAGL Distribution.⁶

1.3 Structure of this document

ActewAGL Distribution's tariff structure for standard control services is set out in chapter 2. The chapter includes details on the components and rationale for each tariff, an outline of ActewAGL Distribution's pricing strategy and how it relates to the pricing principles in the transitional *Rules*, and an explanation of the price setting process—that is, the process of moving from the regulated maximum allowed average revenue and a set of high level pricing principles to a full schedule of tariffs and charges for 2013/14.

The proposed network tariffs and charges for ActewAGL Distribution's standard control services for 2013/14 are presented in chapter 3. The chapter includes discussion of the changes relative to 2012/13.

The structure and basis of ActewAGL Distribution's charges for alternative control (metering) services, the proposed charges for 2013/14 and the changes relative to 2012/13 are presented and explained in chapter 4.

Indicative estimates of the likely impacts of the price changes on average customer electricity bills are provided in chapter 5.

³ AER, *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision*, April 2009

⁴ AER 2009, *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision*, April.

⁵ AER 2009, *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision*, April, p. viii

⁶ AER 2009, *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision*, April, p. viii

2 The structure and basis of ActewAGL Distribution's network tariffs

The transitional *Rules* (clause 6.18.2) require a description of the *tariff classes*⁷ and tariffs that are to apply in 2013/14. For each tariff within a tariff class, the *charging parameters*⁸ and the elements of service to which they relate must also be set out in the pricing proposal.

2.1 Network tariff structure

ActewAGL Distribution's offers network tariffs in three tariff classes:

- Residential;
- Commercial low voltage (LV); and
- High voltage (HV).

The transitional *Rules* stipulate that tariff classes must be constituted with regard to the need to group customers together on an economically efficient basis and the need to avoid unnecessary transactions costs (clause 6.18.3(d)). ActewAGL Distribution meets this requirement by grouping customers according to type of connection (residential or commercial), and connection voltage (LV or HV). Customers within each class have similar load and connection characteristics. The relevant costs for each class can then be identified and reflected in the tariffs for each class.

Within each of these three tariff classes, ActewAGL Distribution has developed a suite of network tariffs that effectively meet the diverse needs of its customer base, encourage efficient use of the network and signal the costs of future network expansion. Residential customers are offered a choice of four network tariff options plus two controlled load off-peak options and an embedded renewable generation tariff option. Commercial LV customers are offered four main tariff options. Commercial customers on the general network charge also have access to the controlled load off-peak tariff options and the embedded renewable generation tariff option on a similar basis to customers in the residential class. Commercial HV customers are offered four tariff options. Customers are able to choose the option which best suits their needs, subject to the eligibility criteria set out in Tables 2.1 to 2.3 below.

The network tariffs comprise different combinations of the following *charging parameters*:

- Network access charges—these apply per customer for residential consumers and per connection point for commercial customers. They involve a fixed daily charge and do not vary with electricity consumption or capacity;
- Energy charges—these apply to each unit of electricity consumed. The c/kWh rate may vary with the level of consumption (with higher rates applying above certain thresholds) or with the time-of-use (with lower rates applying at off-peak periods);
- Maximum demand charges—these apply per connection point for some commercial tariffs. They involve a charge per unit of maximum demand (in c/kVA/day). The

⁷ A tariff class is defined in chapter 10 of the *National Electricity Rules* as “a class of customers for one or more direct control services who are subject to a particular tariff or particular tariffs”.

⁸ Charging parameters are defined as “the constituent elements of a tariff” in chapter 10 of the *National Electricity Rules*.

maximum demand is the highest demand calculated over a 30-minute interval during the billing period;

- Capacity charges—these apply on the same basis as maximum demand charges, but are for the maximum demand calculated over a 30-minute interval during the previous 12 months.

Network access charges relate to the connection services provided to customers. They are associated with the costs of the provision and use of connection assets for each tariff class, as well as customer related costs such as network call centre costs.

Energy charges relate to the distribution services provided to customers. They are linked to the costs of the provision and use of distribution assets (other than connection assets), and also recover most of the common services costs. Higher energy rates at peak periods reflect higher costs. Higher energy rates beyond 330 kWh per day for the general network charge encourage larger consumers with a good load factor to move to demand or time-of-use network charges.

Maximum demand and capacity charges relate to the capacity provided to meet maximum demand and provide incentives for consumers to manage their load on the network.

The allocation of costs to charging parameters is discussed further in section 2.4 below.

The tariffs and charging parameters for each tariff class are shown in the following tables (2.1 to 2.3). The tables include an explanation of the purpose of each tariff and the customers to which each tariff may apply.

2.1.1 Network tariffs for residential customers

ActewAGL Distribution's residential network tariff structure is shown in Table 2-1. The *Residential time-of-use (TOU)*, *Residential 5000* and *Residential with heat pump* tariffs are refinements of the *Residential basic* tariff to reflect customer load profiles. The *Residential TOU* tariff provides an opportunity and an incentive for customers with the necessary metering capability to respond to price signals at different times of the day, where reflected in the final price of their retailer, and manage their electricity bill in line with the costs they impose on the network. The *Residential TOU* tariff is the default tariff for all new residential and commercial connections. The *Residential 5000* and *Residential with heat pump* tariffs involve a higher connection charge and an inclining block structure with a higher energy charge (cents per kWh) applying above certain thresholds. These tariffs more accurately tailor costs to the load profile of these customers. The off-peak tariff options can be used in conjunction with the *Residential basic* and the *Residential TOU* network charges.

Table 2-1 Network tariff structure – residential

Tariff	Charging parameters	Explanation
Residential basic network	Network access charge (c/day/customer) Energy charge (c/kWh)	The residential basic network tariff is available to installations at private dwellings, excluding serviced apartments, but including: • Living quarters for members and staff of religious orders; • Living quarters on farms; • Charitable homes; • Retirement villages; • Residential sections of nursing homes and hospitals; • Churches, buildings or premises which are primarily used for public worship; and • Approved caravan sites. The energy charge varies neither with the level of consumption nor the time of day. However, customers on this tariff are also eligible for the off-peak tariffs.
Residential time-of-use (TOU) network ⁹	Network access charge (c/day/customer) Energy at max times, ie 7am to 9am and 5pm to 8pm every day (c/kWh) Energy at mid times, ie 9am to 5pm and 8pm to 10pm every day (c/kWh) Energy at economy times, ie all other times (c/kWh)	This tariff is available to residential customers (as defined above) with a meter able to be read as a time-of-use meter and to electric vehicle recharge facilities on residential premises. The energy charges relate to the supply of network services at various times. Higher rates apply at max or peak times to encourage users to shift their load to off-peak periods. Customers on this tariff are also eligible for the controlled load off-peak tariffs. Residential consumers with a meter with two registers capable of providing time-of-use consumption data from each register may have the time-of-use charges applied separately to each register.
Residential 5000 network	Network access charge (c/day/customer) Energy for the first 60 kWh/day (c/kWh) Energy above 60 kWh/day (c/kWh)	This tariff is designed for residential customers who have large continuous (rather than time controlled) loads, such as electric hot water systems, and consume over 5,000 kWh per annum. The energy charges relate to the supply of network services above and below certain volume thresholds. An inclining block structure applies (ie higher energy rates for the second block of energy). The lower energy rate is limited to consumption up to 60 kWh per day, reflecting a typical domestic usage profile. This is sufficient to cover the energy requirements of many residential consumers.
Residential with heat pump	Network access charge (c/day/customer) Energy for the first 165 kWh/day (c/kWh) Energy above 165 kWh (c/kWh)	This tariff is only available to residential customers with a reverse cycle air conditioner. An inclining block structure applies (ie higher energy rates for the second block of energy). The lower energy rate is set to recover the incremental cost of energy load on the network as a demand management tool to lower winter peak loads and improve utilisation of the network in summer and so improve overall network utilisation.
Off-peak (1) night network	Energy at controlled times, ie between 10 pm and 7 am (c/kWh)	The off-peak (1) night charge is available only to customers utilising a controlled load element, and taking all other energy at residential basic network, residential time-of-use or general network rates. It is not available where the customer's meter is read as an interval meter. The off-peak (1) night charge is applicable to permanent heat (or cold) storage; electric vehicle

⁹ All times for metering are Eastern Standard Time.

Tariff	Charging parameters	Explanation
		recharge; and CNG vehicle gas compression installations. The design and rating must be acceptable to ActewAGL. The installation must use most energy during the controlled times but may be boosted at the principal charge, or charges, at other times. The off-peak (1) night network energy charge relates to supply of network services at controlled times, for 6 to 8 hours per day between the hours of 10 pm and 7 am.
Off-peak (3) day and night network	Energy at controlled times, ie between 10 pm and 7 am and 9 am and 5 pm (c/kWh)	Available only to customers utilising a controlled load element, and taking all other energy at residential basic network, residential time-of-use or general network rates. This charge is not available where the customer's meter is read as an interval meter. This charge is applicable to permanent heat (or cold) storage installations. The design and rating must be acceptable to ActewAGL. The off-peak (3) day and night network energy rate applies to power supplied for up to 13 hours per day between 10 pm and 7 am and again between 9 am and 5 pm.
Renewable generation	Energy charges (c/kWh)	This tariff applies to customers with grid connected solar or wind energy generation systems. Different arrangements apply to customers participating in the ACT feed-in tariff scheme, in accordance with the <i>Electricity Feed-in (Renewable Energy Premium) Act 2008</i> (ACT). For 2013/14, net metering applies for new PV installations.

2.1.2 Network tariffs for low voltage commercial customers

ActewAGL Distribution sets different tariffs for commercial low voltage (LV) and commercial high voltage (HV) customers recognising the different costs associated with supplying each group. Within the commercial LV tariff class a range of tariff options has been developed to meet the diverse needs of commercial customers and to accommodate their differing load profiles and ability to respond to price signals. Of the four main options offered to commercial LV customers, all but the *General network* tariff involve time-of-use charges. The *General network* tariff does, however, involve an inclining block tariff structure with higher energy charges (c/kWh) applying above certain thresholds. Also, the off-peak (controlled load) tariffs are available to customers on the *General network* tariff.

Two of the commercial LV options involve capacity and/or maximum demand charges, in conjunction with time-of-use charges. Customers able to improve their load factor¹⁰ have an incentive to choose a tariff with a demand or capacity charge, and reduce their energy bills. Customers on the *General network* and *General time-of-use network* charges can move to the demand charges and lower their network costs if they have a sufficiently large load (for the cost savings to offset the higher cost of interval metering) and a reasonable load profile.

¹⁰ The load factor is the ratio of average load to peak load.

Table 2-2 Network tariff structure - commercial low voltage

Tariff	Charging parameters	Explanation
General network	Network access charge (c/day/customer) Energy for the first 330 kWh/day (c/kWh) Energy above 330 kWh/day (c/kWh)	This tariff is available to all customers. The tariff is most suitable for small commercial consumers operating in regular business hours or larger customers with poorer load factors (peaky loads). This tariff may be used in conjunction with the off-peak tariffs.
General TOU network	Network access charge (c/day/customer) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This tariff is particularly suitable for small commercial customers with discretionary or relatively large off-peak loads such as bakers, freezer installations and irrigators. The energy charges relate to supply of network services at different times.
LV TOU kVA demand network	Network access charge (c/day/connection point). Maximum demand (in billing period) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This tariff is appropriate for customers with an average or stable commercial load. The maximum demand charge is designed to encourage consumers to manage their demand upon the network. The energy charges relate to supply of energy at different times, with lower rates in off-peak times reflecting the availability of capacity and encouraging consumers to shift their load from peak to off-peak times to utilise the available capacity.
LV TOU capacity network	Network access charge (c/day/connection point) Maximum demand (in billing period) (c/kVA/day) Capacity (max demand in last year) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This tariff is open to all low voltage consumers and intended to reward those customers with seasonally stable loads. It is prescribed for low voltage customers with embedded generation. The tariff provides an incentive for customers with embedded generation to manage their output and their down-times (eg for servicing) so as to minimise their demand on the network.
Streetlighting	Network access charge (c/day/customer) Energy at any time (c/kWh)	This tariff applies to the night-time lighting of streets and public ways and places.
Small unmetered loads	Network access charge (c/day/customer) Energy at any time (c/kWh)	This tariff applies to eligible installations as determined by ActewAGL, including: • telephone boxes • telecommunication devices • other, as determined by the National Metrology Coordinator. Energy charges are calculated based on the assessed rating of the load and the charge period.
Internal (Discontinued)		This tariff is discontinued. Customers on this tariff are to move to the General Network charge for 2013/14.

* Business times are between 7 am and 5 pm Eastern Standard Time on weekdays. Evening times are between 5 pm and 10 am Eastern Standard Time on weekdays. Off-peak times are all other times.

2.1.3 Network tariffs for high voltage customers

To qualify for the high voltage demand network charges, consumers must take their energy at high voltage (nominal voltage not less than 11 kV) and make a capital contribution towards their connection assets and transformers. High voltage consumers have the option of owning and operating their own high voltage assets. Some customers have aggregated their load, incorporating part of ActewAGL Distribution's low voltage network to become a high voltage customer. A separate high voltage network charge is available for such customers.

Customers taking their energy at high voltage also have the option of selecting the network tariffs available to low voltage consumers. For example, a high voltage customer with a poor load factor could select the *General* or the *General time-of-use* network charge.

Table 2-3 Network tariff structure - high voltage

Tariff (code)	Charging parameters	Explanation
HV TOU Demand Network (111)	Network access charge (c/day/connection point) Max demand (in billing period) (c/kVA/day) Capacity (max demand in past year) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This tariff is appropriate for large customers taking supply at high voltage with a low voltage network owned and maintained by ActewAGL. The energy charges relate to supply of network services at different times, with lower rates in off-peak times encouraging customers to increase their utilisation of the network in off-peak periods. The demand charge is applied to the maximum demand in the billing period while the capacity charge is applied to the maximum demand in the previous 12 months. The capacity charge encourages consumers to monitor and manage their peak demand over the year while the demand charge continues to encourage consumers to manage their demand requirements each month.
HV TOU Demand Network – Customer HV (112)	Network access charge (c/day/connection point) Max demand (in billing period) (c/kVA/day) Capacity (max demand in past year) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This tariff is appropriate for large customers taking supply at high voltage with a low voltage network owned and maintained by ActewAGL, where the customer owns and is responsible for their high voltage assets (including transformers and switch gear).
HV TOU Demand Network – Customer LV (121)	Network access charge (c/day/connection point) Max demand (in billing period) (c/kVA/day) Capacity (max demand in past year) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This network charge is appropriate for large customers taking supply at high voltage where the customer owns and is fully responsible for their own low voltage network. The network access charge relates to the connection services provided to the customer including provision of the current transformer necessary to meter these large loads. The capacity charge encourages the consumer to monitor and manage their peak demand over the year while the demand charge continues to encourage consumers to manage their capacity requirements each month. The energy charges relate to supply of network services at different times, with lower rates in off-peak times reflecting the relatively low costs of off-peak supply, and thereby providing incentives for customers to switch their utilisation of the network to off-peak periods.
HV TOU Demand Network – Customer HV and LV (122)	Network access charge (c/day/connection point) Max demand (in billing period) (c/kVA/day) Capacity (max demand in past year) (c/kVA/day) Energy at business times* (c/kWh) Energy at evening times (c/kWh) Energy at off-peak times (c/kWh)	This network charge is appropriate for large customers taking supply at high voltage where the customer owns and is fully responsible for their own low voltage network and where the customer owns and is responsible for their high voltage assets (including transformers and switch gear).

* Business times are between 7 am and 5 pm Eastern Standard Time on weekdays. Evening times are between 5 pm and 10 am Eastern Standard Time on weekdays. Off-peak times are all other times.

2.1.4 Miscellaneous charges and fees

In addition to the network tariffs set out above, ActewAGL Distribution offers a range of miscellaneous services with charges for each service. These services are part of ActewAGL Distribution's standard control services, as set out in Appendix D of the Final Distribution Determination. The charges are regulated through the maximum allowed average revenue cap applying to standard control services. The structure of each charge depends on the type of service. Some services are charged on a per visit basis, others per installation or per test. The charges for miscellaneous services are set on a cost reflective basis. For example, separate rates apply for temporary connections depending on whether they relate to an overhead, standard underground or free-standing underground connection, as these will involve different costs. Standard control miscellaneous services fees and charges are discussed further in chapter 3.

ActewAGL Distribution also provides some miscellaneous services as part of its alternative control (metering) services. These include meter tests, installation of interval meters at a customer's request, and special meter reads. These are covered in chapter 4.

2.2 Pricing strategy

ActewAGL Distribution has developed and refined its network tariff structure over time, guided by its pricing strategy. The strategy involves:

- Setting prices to signal to customers the economic costs of providing distribution services;
- Providing customers with a choice of flexible and innovative tariffs to best meet their needs;
- Providing incentives and opportunities for demand management;
- Ensuring that tariffs are set to recover costs in a way that encourages efficient use of the network and signals to customers the cost of network expansion; and,
- Offering customers a clear and simple tariff structure, noting the need to take account of the ability of different customer groups to respond to price signals and the need to keep transactions costs low.

ActewAGL Distribution's pricing strategy has in recent years accommodated the development of some innovative tariffs and significant customer responses. For example, in line with the strategies of setting cost reflective prices and providing opportunities and incentives for demand management, ActewAGL Distribution has gradually introduced several time-of-use charging options for both commercial and residential customers. More than 50 per cent of the total load in the ACT is now subject to time-of-use or controlled load (off-peak) charges. For the non-residential sector, nearly 80 per cent of the load is on time-of-use or controlled load tariffs.

In October 2010, time-of-use tariffs became the default tariff for all new residential and commercial premises, with the option to select an alternative tariff.

In the 2012 *Power of Choice* Final Report, the AEMC recommended "a gradual phase-in of efficient and flexible retail pricing options for residential and small business consumers

through the introduction of cost reflective electricity distribution network pricing structures”.¹¹ ActewAGL Distribution supports the gradual introduction of efficient and flexible tariff options, and has been implementing these in recent years. However, ActewAGL Distribution does not support the AEMC’s proposed approach to the phase-in, which would involve a shift away from the current approach of setting TOU tariffs as the default for all new customers and giving all customers choice.¹²

The application of maximum demand and capacity charges in several commercial tariff options has further strengthened price signals to customers, provided incentives to use the network more efficiently and resulted in significant customer response. The maximum demand charges signal to customers the relatively high cost of providing capacity to meet demand and provide incentives to customers to improve both their load factor (that is, spread their load more evenly) and power factor (which allows the existing network to deliver more energy). Between 1999/00 and 2011/12, customers on the *Low voltage demand* network tariff improved their load factor and therefore their utilisation of the network by 12.5 per cent, increasing the average energy consumed relative to the average of their monthly maximum demand from 40.1 per cent to 45.1 per cent. Over the same period, high voltage customers increased their load factor, and therefore their utilisation of the network, from 54.2 per cent to 61.0 per cent, an improvement of 12.6 per cent.

These price signals have been effective demand management tools and have allowed ActewAGL Distribution to keep network augmentation costs to a minimum. For example, until the current regulatory period, no new zone substations had been built in the ACT since 1994.

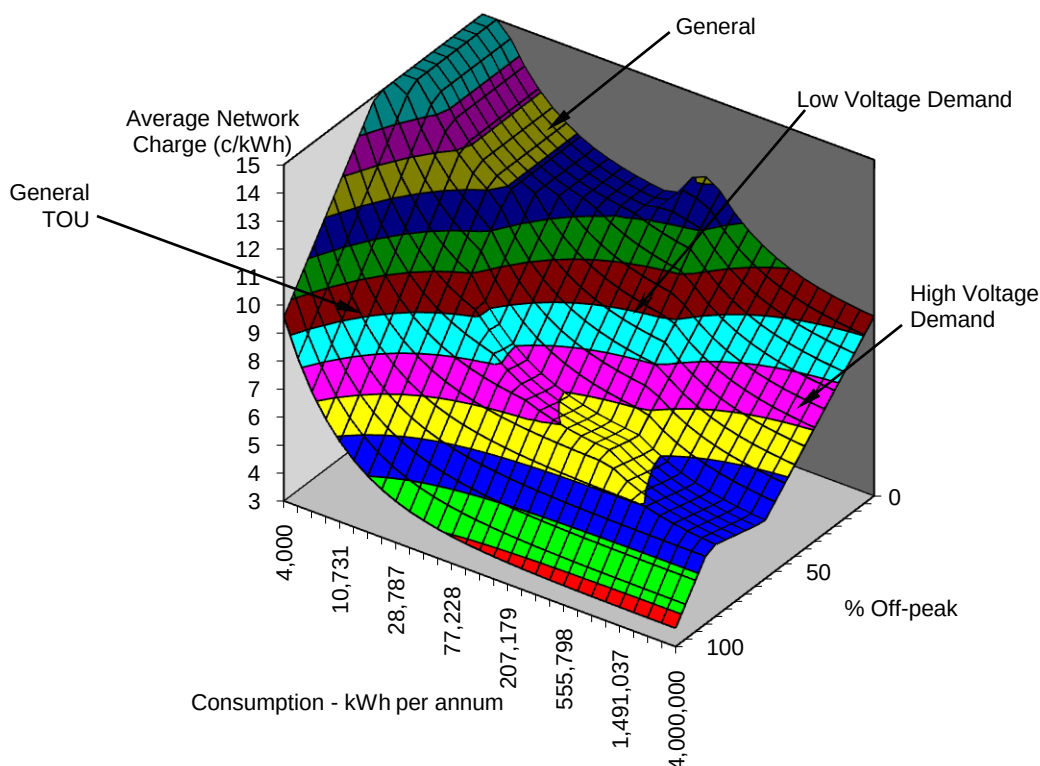
The options available for commercial customers and the incentives created are represented in Figure 2.1 which shows the main network tariffs available to commercial customers, including time-of-use and demand tariffs. Sophisticated modelling ensures that the tariffs are synchronised and that the costs of supplying particular loads are appropriately priced. ActewAGL Distribution is able to offer a suite of commercial tariffs and be indifferent to customers’ choices between them.

The graph plots the average minimum network charges over a range of consumption levels and a range of consumption profiles from all *peak time* (business and evening) to all *off-peak* for commercial customers. It shows how the *General network* charge suits small commercial customers with relatively large peak time loads, while those with high off-peak loads can be rewarded by selecting the *General time-of-use network* tariff. The graph also demonstrates how the low voltage and high voltage demand tariffs align with the non-demand tariffs. Larger customers with good load profiles are able to choose a demand based tariff option and reduce overall network supply costs and their average network charges.

¹¹ AEMC 2012, *Power of choice review: giving consumers options in the way they use electricity, Final Report*, recommendation 14, p. ii

¹² The AEMC’s recommended phase-in involves: mandatory “flexible pricing” for “large load customers”; opt-out (ie transition to flexible network tariffs with option to revert to flat tariffs) for “medium to large load customers”; and opt-in (ie option to adopt flexible tariffs) for “small to medium load customers”. See AEMC 2012, op cit, p. 162

Figure 2.1 Average network charges for commercial customers



2.3 Consistency with the transitional Rules pricing principles

The transitional *Rules* include pricing principles to be applied when setting tariffs (clause 6.18.5). These principles recognise the complexities which arise in electricity network pricing because of the nature of electricity distribution costs—in particular the large fixed costs associated with providing capacity to meet peak demand, the high degree of shared or joint costs, and costs which vary according to the time of day and season and voltage of supply. There is no unique set of efficient and equitable prices, but rather a range within which prices should fall, taking account of the need to raise sufficient revenue to cover costs, the need to signal to customers the costs of expanding capacity to meet future demand, and the need to take account of the scope for customers to respond to price signals, bearing in mind the administrative and technical costs of doing so.

Clause 6.18.5(a) of the transitional *Rules* requires that for each tariff class the revenue must lie between an upper bound of the *stand alone costs* and a lower bound of *avoidable cost*. The stand alone cost for any group of customers is the cost that would be incurred if only that customer group were supplied. Any costs that would otherwise be shared with other customer groups would have to be fully attributed to the stand alone customers. The stand alone cost is

effectively the cost of replicating or bypassing the infrastructure.¹³ The avoidable cost for any group of customers is the cost that would be avoided if the group of customers were removed from the network.

The purpose of the *between stand alone and avoidable cost* test is to ensure that there are no cross subsidies between tariff classes. Compliance with the test discourages inefficient bypass of the network (which may occur if prices are above the stand alone cost).¹⁴ If any tariff class, or group of customers, is paying less than the avoidable cost lower bound—the costs that arise directly from its use of the network—then it is receiving an economic subsidy. If any tariff class is paying more than the stand alone cost upper bound—the cost if it alone were supplied—then it is subsidising other users. If the revenues from each group lie between the bounds of avoidable cost and stand alone cost then each group is making a contribution to covering the joint or shared costs of supply, while no group is receiving or paying an economic subsidy.

Clause 6.18.5(b) of the transitional *Rules* says that each tariff and charging parameter must take account of the long run marginal cost of the service (or element of the service) to which it relates, while also taking account of transactions costs and whether customers are able or likely to respond to price signals. The purpose of the long run marginal cost requirement is to ensure that prices signal to customers the forward-looking costs of meeting additional demand or the savings from reduced demand.

Clause 6.18.5(c) of the transitional *Rules* says that if, as a result of the operation of clause 6.18.5(b), expected revenue may not be covered, tariffs must be adjusted in a way that results in minimum distortion to efficient patterns of consumption.

ActewAGL Distribution's pricing strategy is consistent with each of the pricing principles in clause 6.18.5 of the transitional *Rules*. As far as possible, costs are allocated to the customer group or groups which are directly responsible for the costs. For example, the costs associated with providing low voltage connection assets are directly attributed to low voltage customers, but not high voltage customers. As a result, the revenue from each tariff class recovers at least the avoidable cost (or directly attributed cost) associated with providing the service to that class. Where additional costs must be recovered, each class pays a share, but none pays more than its stand alone cost. In this way, the requirement of providing subsidy-free prices is met. This is demonstrated in section 2.3.1 below, where revenue from each tariff class is shown to lie between the relevant bounds of avoidable and stand alone costs.

The principle of signalling to users the long run marginal cost is also taken into account in ActewAGL Distribution's approach to pricing. The long run marginal cost is the additional cost of meeting additional future demand. It includes the capital costs associated with any increment in capacity required to meet the additional demand, as well as the additional operating and maintenance costs. At times when demand is at or near full network capacity, prices should signal the costs of expanding capacity rather than the short run marginal costs which do not include capital related costs.

¹³ NERA 2006, *Distribution Pricing Rule Framework, Report prepared for the Network Policy Working Group of the Standing Committee of Officials (SCO) of the Ministerial Council on Energy (MCE)*, December, p. 22

¹⁴ NERA 2006, *Distribution Pricing Rule Framework*, December, p. 22

ActewAGL Distribution's capacity and maximum demand charges provide signals to commercial customers about the cost of capacity expansion and, therefore, encourage these users to manage their maximum demands. The *General time-of-use* tariff does not incorporate a demand charge, but it does signal to customers the relatively high cost of providing capacity at peak times.

While residential tariffs do not have capacity or maximum demand charges, a TOU tariff is available and off-peak tariff options can be combined with other tariffs (including the *Residential basic* tariff). The higher energy rates at peak periods signal to customers the cost of expanding capacity and, therefore, encourage consumers to shift their load to off-peak periods. The residential time-of-use tariff is now the default tariff for new residential consumers.

When setting tariffs to ensure that all costs are recovered, ActewAGL Distribution also aims to minimise the distortion to efficient patterns of consumption. This influences the balance between the network access charges (which do not vary with the level of consumption) and the energy charges (which vary with the level of consumption and therefore more directly influence consumption patterns). For example, ActewAGL Distribution seeks to minimise the fixed charges in the *General time-of-use* and the *Low voltage demand* tariffs to ensure that small customers are not inefficiently discouraged from adopting the tariffs with time-of-use and demand price signals. Also, for residential customers, ActewAGL Distribution has kept the fixed charge (including daily metering charge) for the *Residential time-of-use* charge the same as that for the *Basic residential* charge to ensure there is not an inefficient barrier to customers seeking to move to a tariff that provides time-of-use price signals.

The practical application of these strategies and principles in the price setting process is described in section 2.4 below.

2.3.1 Calculating avoidable, stand alone and long run marginal costs

ActewAGL Distribution engaged economic consultants NERA in 2009 to advise on the application of the pricing principles and to provide estimates of avoidable costs, stand alone costs and long run marginal costs. NERA undertook a detailed analysis of the available methods and developed and implemented a recommended approach to calculating avoidable, stand alone and long run marginal costs

ActewAGL Distribution has used NERA's estimates of avoidable cost for 2009/10 for each tariff class escalated in line with the increase in revenue for each customer class. These costs have been used to calculate the stand alone costs for each tariff class using NERA's methodology. The results for avoidable and stand alone costs are shown in Table 2.4. The table also shows that average 2013/14 DUOS revenue for each tariff class lies within the lower bound of avoidable cost and the upper bound of stand alone costs. The tariffs therefore comply with the requirement in clause 6.18.5(a) of the transitional *Rules*.

Table 2-4 Avoidable and stand alone costs 2013/14 (\$'000)

<i>Tariff class</i>	<i>Avoidable cost ('000)</i>	<i>DUOS charges ('000)</i>	<i>Stand alone cost ('000)</i>
Residential	\$11,124	\$72,376	\$185,483
Commercial low voltage	\$1,392	\$99,803	\$175,751
High voltage	\$62	\$14,757	\$174,421
Total		\$186,936	

NERA's analysis of long run marginal costs highlights a range of practical and theoretical issues that need to be addressed in the calculation and interpretation of long run marginal cost, particularly where the distribution network may not be in a 'steady-state'.¹⁵ ActewAGL Distribution considers that these matters should be carefully examined in the context of the AER's development of guidelines for the calculation of stand alone, avoidable and long run marginal costs (in accordance with clause 6.2.8 of the transitional *Rules*).

Following a detailed review of available methods and methods used in practice, NERA developed for ActewAGL Distribution a long run marginal cost model based on the average incremental cost (AIC) approach. Recognising that the long run marginal cost of network expansion varies depending on the network type (that is, voltage level), NERA estimated the long run marginal cost for each network type then translated this into a cost estimate for each customer class on the basis of the contribution of each class to demand at times of network peak demand.¹⁶

NERA's estimates of long run marginal cost by network level, on a \$/kVA per year basis, are shown in Table 2.5. The estimates are expressed separately for each network level, with the low voltage estimate broken down to reflect the long run marginal cost of network assets used by low voltage commercial customers, those used by low voltage residential customers and shared low voltage assets.

Table 2-5 Long run marginal costs per network level (\$/kVA pa)

<i>Network level</i>	<i>Long run marginal cost \$/kVA pa (\$2010)</i>
Low voltage residential	239.57
Low voltage commercial	29.76
Low voltage shared	4.74
High voltage	113.91

NERA's estimates of long run marginal cost for each tariff class are shown in Table 2-6. These long run marginal cost estimates represent the costs of meeting an additional kVA of demand for each of the three tariff classes:

¹⁵ For example, while the Turvey approach is generally considered to have the stronger theoretical basis, the Average Incremental Cost (AIC) method is more widely used as it is easier to apply. However, the AIC method has important shortcomings in that it is based on average capital costs associated with an increment in demand, not marginal, and the results are also sensitive to the demand forecasts used and the time frame for the analysis.

¹⁶ NERA 2010, *Analysis of ActewAGL's Electricity Distribution Services Costs, A report for ActewAGL*, commercial-in-confidence, April, p. 11

- For a customer located in the HV network, the long run marginal cost reflects the cost of augmenting the high voltage network, in order to meet that additional demand;
- For a low voltage commercial customer, the long run marginal cost reflects the cost of augmenting those areas of the low voltage network used exclusively by those customers, plus the costs of augmenting those areas of the low voltage network which are shared between low voltage commercial and residential customers and plus the costs of augmenting the high voltage network;
- Similarly, the long run marginal cost of meeting an increase in residential customer demand reflects the costs of augmenting those areas of the network used exclusively by residential customers plus those parts of the low voltage network which are shared between low voltage commercial and residential customers and plus the costs of augmenting the high voltage network.

As a result of applying this method of allocating costs, the long run marginal cost estimates for the low voltage customer classes are above the long run marginal cost estimates for the high voltage customer class.

Table 2-6 Long run marginal costs by tariff class (\$/kVA pa)

<i>Tariff class</i>	<i>Long run marginal cost \$/kVA pa (\$2010)</i>
Residential	367.06
Commercial low voltage	157.25
High voltage	113.91

ActewAGL Distribution notes that the long run marginal cost estimates are significantly influenced by a number of data limitations as well as features of the ACT demand and expenditure profiles in the current regulatory period. These features reflect ActewAGL Distribution's stage in the investment cycle—that the distribution network is not in a steady-state—and the lumpy nature of network investment. Care must therefore be taken when interpreting the results and drawing implications for tariff design.

For example, some of the expected future increase in electricity demand is associated with extensions of ActewAGL Distribution's existing distribution network to new residential subdivisions. NERA explains that, conceptually, the capital expenditure associated with extending the current network to meet additional demand in new areas can be distinguished from the capital expenditure associated with meeting increments in demand on the existing network. It is the latter which should be reflected in the estimate of long run marginal cost, as this provides the relevant behavioural signal to consumers. However in practice it is difficult to isolate incremental demand in existing network areas from demand forecasts for ActewAGL Distribution's overall network. As a result NERA calculated the long run marginal cost on the basis of total additional demand and total capital augmentation, including extension of the network to new areas.¹⁷

¹⁷ NERA 2010, *Analysis of ActewAGL's Electricity Distribution Services Costs, A report for ActewAGL*, commercial-in-confidence, April, p. 20

The unique circumstances in the current regulatory period, involving a significant increase in ActewAGL Distribution's forecast total capital augmentation costs, together with very low forecast growth rates for residential demand, (attributable to the declining average residential consumption) result in proportionately higher estimates of long run marginal cost for residential customers. For these reasons, and consistent with the obligation to take long run marginal cost into account, the marginal cost estimate must be carefully interpreted and refined to reflect actual capacity availability in existing residential areas, otherwise it would inadvertently provide a signal to existing customers that there were capacity constraints when the existing residential network has sufficient capacity to meet demand.

2.4 The price setting process

The process of moving from the maximum allowed average revenue (MAAR) as set by the AER to a set of network tariffs involves the following steps:

1. Determine the maximum revenue to be recovered through distribution use of system (DUOS) charges. This requires some adjustments to the MAAR set by the AER in the Final Distribution Determination, as described in section 2.4.1 below.
2. Allocate the total revenue requirement to cost pools, taking account of the cost drivers (for example, whether the costs are demand related or customer related) and the type of assets involved (high voltage or low voltage). This cost allocation process is described in section 2.4.2 below.
3. Allocate the costs to tariff classes via a combination of network access charges, energy charges, demand charges and capacity charges. Costs are allocated according to the type of connection (residential, commercial, LV or HV) and the load profile. This process of setting the distribution use of system (DUOS) charges for each tariff class is described in section 2.4.3 below.
4. Allocate transmission use of system (TUOS) charges to tariff classes, to determine the total charges (DUOS + TUOS) to apply for each tariff class. The process of allocating TUOS charges is described in section 2.4.4 below.

2.4.1 Revenue to be recovered through DUOS charges

In accordance with the AER's Final Distribution Determination (chapter 4, section 4.5) and guideline for standard control services (appendix B, table 1), the maximum allowed average revenue for 2013/14 (in c/kWh) is converted to a total allowed revenue for 2013/14 by multiplying by 2012 throughput (in kWh). Approved cost pass-throughs (positive or negative) are then added to this allowed revenue. Miscellaneous services charges are deducted, since, although part of the MAAR for standard control services, they are set separately as described above. The resulting value is the total maximum amount to be recovered through DUOS charges.

The relevant values for each of these components and the calculation of the DOUS cap for 2013/14 are provided in chapter 3 of this document.

2.4.2 Allocation to cost pools

The total amount to be recovered in 2013/14 is allocated to three cost pools:

- Connection service costs
- Demand related service costs
- Common services costs

The main component of *connection service costs* is the cost of providing connection services to customers—that is, the cost associated with service mains (the wires that connect the customer’s premises to the network).¹⁸ The cost pool also includes other costs that are related to the number of customers or connections, for example network call centre costs.

The connection service costs are allocated across tariff classes using the proportion of connections in each tariff class relative to the total number of connections.

The main components of *demand related service costs* are the costs of reticulation and providing capacity. The split between HV and LV costs is important as it ensures that LV and HV customers are each allocated the appropriate costs—for example, HV customers without any ActewAGL LV reticulation are not required to pay for low voltage mains reticulation. This cost pool includes some shared costs which are demand-related, or related to the size of the network (for example, training and apprenticeships costs), but which cannot be attributed to certain voltages.

The demand related service costs are allocated across tariff classes on a cost reflective basis using energy consumed at different parts of the load cycle (business, evening and off-peak times) based on the method of intercepts.

Common services costs are shared across all users on the basis of energy consumption.

2.4.3 Allocating costs to tariff charging parameters

The cost of connection services and other customer related costs are generally recovered through the network access charges. Demand related service costs are recovered through the energy, maximum demand and capacity charging parameters for those commercial tariffs where these components apply. Costs for common services are recovered in energy charges.

The allocation from cost pools to charging parameters is summarised in Table 2-7.

¹⁸ Metering assets and costs are regulated separately under the alternative control services control mechanism.

Table 2-7 Allocating service costs to tariff charging parameters

	Cost pools		
	Connection service costs	Demand related service costs	Common service costs
Residential			
Network access charge	Residential connection service costs and other customer related costs are recovered through network access charges.	For the <i>Residential 5000</i> and <i>Residential with heat pump</i> network charges, some demand related service costs are recovered via network access charges.	
Energy charges		Energy demand related service costs for residential customers are generally recovered through residential energy charges.	Common service costs are recovered in energy charges.
Commercial LV			
Network access charge	Commercial connection service costs and other customer related costs are recovered through network access charges.		
Energy charges		Demand related costs are recovered through the energy charges.	Common service costs are recovered through the energy charges.
Capacity and maximum demand charges		Where demand and capacity charges apply, demand related service costs are mostly recovered in demand and capacity charges.	
Commercial HV			
Network access charge	High voltage connection service costs and other customer related costs are recovered through network access charges		
Energy charges		Some of the demand related costs are recovered through the energy charges.	Common service costs are recovered through the energy charges.
Capacity and maximum demand charges		Most demand related costs are recovered through the demand and capacity charges.	

2.4.4 Allocating transmission use of system charges

Approximately 62 per cent of TransGrid's TUOS charges to the ACT are on a postage stamp basis—only 38 per cent are on a cost reflective basis. TransGrid's transmission charges comprise a fixed charge, an energy charge (with general and common service components) and a monthly maximum demand charge. The fixed transmission exit charge is about 1.0 per cent of the total and varies according to the size of the connection. ActewAGL Distribution recovers this component of the transmission charge in its network energy charges or its demand and capacity charges (where possible) thereby ensuring that it is allocated according to customers' utilisation of the transmission network. The transmission energy charges are also allocated to network energy charges. For commercial tariffs that include a maximum demand charge, the demand component of the transmission charge is allocated to the maximum demand charge. In all other cases TransGrid's maximum demand charge is allocated to the network energy charges.

3 Network tariffs for 2013/14

3.1 The average revenue cap for standard control services

3.1.1 Base maximum allowable average revenue for standard control services

ActewAGL Distribution's standard control services prices are regulated using a maximum allowable average revenue (MAAR) cap. To calculate the maximum allowable average revenue for 2013/14, the following formula is used, as required by the AER's Final Distribution Determination:¹⁹

- Maximum allowable average revenue per kWh (MAAR_t) = MAAR_{t-1} × (1+CPI_t) × (1- X_t)

where:

- MAAR_{t-1} is the maximum allowable average revenue per kWh in the previous year; and
- CPI means the Consumer Price Index, All Groups Index number for the weighted average of eight capital cities.

The CPI figure will be determined using the following formula:

$$CPI_{(t)} = \frac{CPI_{March(t-2)} + CPI_{June(t-2)} + CPI_{Sept(t-1)} + CPI_{Dec(t-1)}}{CPI_{March(t-3)} + CPI_{June(t-3)} + CPI_{Sept(t-2)} + CPI_{Dec(t-2)}} - 1$$

where:

- year t is the year for which tariffs are being set; and
- year t – 1 is the previous regulatory year.

The CPI adjustment figure for the year commencing 1 July 2013 is calculated as follows:

$$CPI_{(2013)} = \frac{CPI_{March(2012)} + CPI_{June(2012)} + CPI_{Sept(2012)} + CPI_{Dec(2012)}}{CPI_{March(2011)} + CPI_{June(2011)} + CPI_{Sept(2011)} + CPI_{Dec(2011)}} - 1$$

Therefore:

$$CPI_{(2013)} = \frac{99.9 + 100.4 + 101.8 + 102.0}{98.3 + 99.2 + 99.8 + 99.8} - 1$$

$$CPI_{(2013)} = \frac{404.1}{397.1} - 1$$

$$CPI_{(2013)} = 1.76\%$$

The maximum average revenue for 2012/13 calculated according to the AER's price determination for that year is \$0.05999 per kWh. For 2009/10 to 2011/12, the X factors are as stated in the AER's Final Distribution Determination. For 2012/13 and 2013/14 the corrected X factor of -5.41 per cent is used, consistent with the AER's decision to revoke and substitute

¹⁹ AER 2009, *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision*, April, p. 18

the determination, in accordance with clause 6.13 of the transitional Rules. The CPI of 1.76 per cent is applied to the maximum allowed average revenue (MAAR) for 2012/13 to obtain the maximum allowed average revenue in 2013/14. The calculations of the MAAR are shown in Table 3.1.

Table 3-1 Calculation of the Maximum Allowable Average Revenue 2013/14

	MAAR previous year	X Factor	Sum of CPI indices	CPI	MAAR
2008/09			631.8		\$0.04092
2009/10	\$0.04092	-13.82%	659.3	4.35%	\$0.04860
2010/11	\$0.04860	-4.00%	671.3	1.82%	\$0.05147
2011/12	\$0.05147	-4.00%	690.4	2.85%	\$0.05505
2012/13	\$0.05505	-5.41%	713.8	3.39%	\$0.05999
2012/13		New CPI base	397.1		
2013/14	\$0.05999	-5.41%	404.1	1.76%	\$0.06435

Note that, while the CPI is shown as a percentage to 2 decimal places, the actual CPI figure applied the MAAR of the previous year is calculated based on the sum of the CPI indices for each year divided by the sum of the CPI indices for the previous year. Also, while the MAAR for 2008/09 is rounded to 5 decimal places, the MAAR figures for subsequent years are not rounded.

The MAAR is applied to the actual energy transported in the previous calendar year to establish a price cap for the following financial year. Therefore, the prices for 2013/14 are based upon energy transported in 2012. The actual energy transported in the 2012 calendar year was 2,909,291,502 kWh. When this is applied to the maximum allowable average revenue for 2013/14 of \$0.06435 per kWh, it results in the revenue cap for standard control services delivered in 2012 of \$187,222,326 which forms the regulated revenue cap for calculating 2013/14 prices.

3.1.2 Recovery of passed through costs in DUOS

In January 2013, the AER approved²⁰ a negative pass through amount of \$727,564 in ActewAGL's 2013/14 distribution charges with respect to the Feed-in Tariff. Also, in January 2013, the AER determined²¹ that ActewAGL could pass through an amount of \$1,997,929.42 for the NECF. Combining the negative pass through amount for the feed-in tariff and the positive pass through amount for the NECF means that the total pass through costs for 2013/14 amount to \$1,270,365. The NECF costs have been allocated to customers using the established methodology for customer related costs.

As explained above, DUOS prices for 2013/14 are calculated based upon the actual load in the 2012 calendar year. The amount of electricity consumed in the ACT has been declining since about July 2011. ActewAGL forecasts that this decline will continue into 2013/14. Factors contributing to this decline include the retail price increase due to the carbon tax and

²⁰ AER. 2013, *Determination, ActewAGL Distribution, Pass through application Australian Capital Territory Feed-in Tariff Scheme 2011-12, January*.

²¹ AER. 2013, *Final Determination, ActewAGL Distribution, Cost pass through application for the National Energy Customer Framework, January*.

the ACT Government's energy efficiency program which is directly aimed at reducing household energy consumption. ActewAGL's latest forecast for 2013/14 aligns with the forecast in the AER's Final Decision. Therefore, it has used the AER forecast of 2,889 GWh. This is 0.7 per cent lower than the 2,909 GWh of energy consumed in 2012.

To recover \$1,270,365 of costs from the 0.7 per cent lower consumption in 2013/14, ActewAGL must adjust upward by 0.7 per cent the amount passed through in network prices calculated using the 2012 throughput. Therefore, the amount of the pass through used to set prices has been inflated to \$1,279,314.

3.1.3 Calculation of the revenue cap for DUOS

The calculation of the revenue to be recovered from 2013/14 distribution use of system (DUOS) charges is shown in Table 3-2.

Table 3-2 Calculation of the revenue cap for DUOS 2013/14

Maximum allowable average revenue 2013/14 (\$/kWh)*	A	\$0.06435
Energy sales ACT 2012 (kWh)	B	2,909,291,502
Standard control services revenue cap	$C = A \times B$	\$187,222,326
Approved pass-throughs/Feed-in tariff adjustment	D	\$1,279,314
Total standard control services revenue cap	$E = C + D$	\$188,501,639
Less miscellaneous fees and charges	F	\$1,215,623
Distribution use of system revenue cap	$G = E - F$	\$187,286,017

* The MAAR shown in this table has been rounded to 5 decimal places. The calculations have been made without rounding.

3.2 Distribution use of system charges

ActewAGL Distribution's proposed DUOS prices for 2013/14 are shown in Table 3-3. These would have recovered \$187,286,003 on the actual customer, demand and energy quantities recorded in the 2012 calendar year. The table shows the tariff classes that are to apply in 2013/14 and the tariffs for each tariff class. The charging parameters for each tariff are set out together with the service to which that charging parameter relates. All prices exclude GST unless otherwise stated. The 2013/14 notional revenue numbers shown in the table are the proposed charges multiplied by the relevant quantities for the previous calendar year (2012).

To demonstrate compliance with the AER's control mechanism, ActewAGL Distribution is required to demonstrate that the sum of the standard control services revenue using the prices for the pricing year and the quantities for the previous calendar year divided by the quantity of energy in kWh transported over the previous calendar year is less than or equal to the MAAR for the pricing year.²²

The sum of DUOS charges, less the pass-through adjustments and miscellaneous fees is \$187,222,312. When this is divided by the 2012 calendar year energy transported in the ACT of 2,909,291,502 kWh, it results in an average price of \$0.06435 per kWh. As this average

²² AER 2009, *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision*, April, p. 18

price is equal to the MAAR, the prices comply with the AER's corrected Distribution Determination.

Table 3-3 Distribution use of system charges 2013/14

<i>Tariff & element of service</i>	<i>Pricing Unit</i>	<i>2012 kWh/ cust no/ kVA</i>	<i>Proposed Charges 2013/14</i>	<i>Notional revenue 2013/14</i>
Residential tariffs				
010 Residential Basic Network				
Network access	c/day/customer	130,167	21.44	\$10,214,282
Energy at any time	c/kWh	923,491,037	5.649	\$52,168,009
015 Residential TOU Network				
Network access	c/day/customer	7,325	21.44	\$574,822
Energy at max times	c/kWh	10,050,658	9.894	\$994,412
Energy at mid times	c/kWh	13,884,998	5.511	\$765,202
Energy at economy times	c/kWh	10,761,240	4.042	\$434,969
020 Residential 5000 Network				
Network access	c/day/customer	8,225	42.64	\$1,283,607
Energy for the first 60 kWh per day	c/kWh	45,724,339	4.109	\$1,878,813
Energy above 60 kWh per day	c/kWh	1,156,844	5.649	\$65,350
030 Residential with Heat Pump Network				
Network access	c/day/customer	4,536	85.34	\$1,416,752
Energy for the first 165 kWh per day	c/kWh	80,019,029	2.893	\$2,314,951
Energy above 165 kWh per day	c/kWh	1,168,768	5.649	\$66,024
060 Off-Peak (1) Night Network				
Energy at controlled times	c/kWh	14,884,138	0.085	\$12,652
070 Off-Peak (3) Day & Night Network				
Energy at controlled times	c/kWh	91,278,494	0.204	\$186,208
Renewable Energy Generation				
Energy consumption (Obsolete)	c/kWh	3,208,260	0.000	\$0
Energy exports	c/kWh	0	0.000	\$0
Commercial low voltage tariffs				
040 General Network				
Network access	c/day/customer	11,987	42.33	\$1,857,179
Energy for the first 330 kWh per day	c/kWh	263,694,449	9.097	\$23,988,284
Energy above 330 kWh per day	c/kWh	25,852,797	12.818	\$3,313,812
135 Small Unmetered Loads Network				
Network access	c/day/customer	23	35.80	\$3,014
Energy at any time	c/kWh	1,513,183	10.738	\$162,489
141 Internal Network (Discontinued- move to General Network)				
Network access	c/day/customer	96	42.33	\$14,854
Energy at any time	c/kWh	798,363	9.097	\$72,627
080 Streetlighting Network				
Network access	c/day/customer	17	43.00	\$2,695
Energy for night time lighting of streets public ways & places	c/kWh	41,722,136	6.126	\$2,555,898
090 General TOU Network				
Network access	c/day/customer	662	42.33	\$102,548
Energy at business times	c/kWh	25,775,332	17.134	\$4,416,345
Energy at evening times	c/kWh	11,684,169	8.532	\$996,893

<i>Tariff & element of service</i>	<i>Pricing Unit</i>	<i>2012 kWh/ cust no/ kVA</i>	<i>Proposed Charges 2013/14</i>	<i>Notional revenue 2013/14</i>
Energy at off-peak times	c/kWh	29,790,219	3.478	\$1,036,104
Low voltage time of use demand network				
101 LV TOU kVA Demand Network				
Network access	c/day/connection point	1,741	49.00	\$312,234
Maximum demand	c/kVA/day	233,561	39.021	\$33,356,519
Energy at business times	c/kWh	372,478,364	4.733	\$17,629,401
Energy at evening times	c/kWh	132,328,674	2.798	\$3,702,556
Energy at off-peak times	c/kWh	404,991,258	1.045	\$4,232,159
103 LV TOU Capacity Network				
Network access	c/day/connection point	25	49.00	\$4,521
Maximum demand (in billing period)	c/kVA/day	7,227	18.082	\$478,278
Capacity (maximum demand in last year)	c/kVA/day	8,324	18.082	\$550,902
Energy at business times	c/kWh	13,606,712	4.733	\$644,006
Energy at evening times	c/kWh	5,502,140	2.798	\$153,950
Energy at off-peak times	c/kWh	18,858,594	1.143	\$215,554
High voltage tariffs				
High voltage time of use demand network with ActewAGL low voltage network				
111 HV TOU Demand Network				
Network access	\$/day/connection point	1	17.00	\$6,222
Maximum demand (in billing period)	c/kVA/day	1,605	14.915	\$87,629
Capacity (maximum demand in last year)	c/kVA/day	1,923	14.492	\$102,024
Energy at business times	c/kWh	2,833,657	3.583	\$101,530
Energy at evening times	c/kWh	1,163,428	2.141	\$24,909
Energy at off-peak times	c/kWh	3,493,430	0.790	\$27,598
112 HV TOU Demand Network – Customer HV				
Network access	\$/day/connection point	0	17.00	\$0
Maximum demand (in billing period)	c/kVA/day	0	13.915	\$0
Capacity (maximum demand in last year)	c/kVA/day	0	13.492	\$0
Energy at business times	c/kWh	0	3.583	\$0
Energy at evening times	c/kWh	0	2.141	\$0
Energy at off-peak times	c/kWh	0	0.790	\$0
High voltage time of use demand network without ActewAGL low voltage network				
121 HV TOU Demand Network – Customer LV				
Network access	\$/day/connection point	22	17.00	\$138,185
Maximum demand (in billing period)	c/kVA/day	67,781	14.915	\$3,700,077
Capacity (maximum demand in last year)	c/kVA/day	80,853	14.915	\$4,413,659
Energy at business times	c/kWh	135,284,203	3.303	\$4,468,437
Energy at evening times	c/kWh	51,880,958	1.791	\$929,188
Energy at off-peak times	c/kWh	170,411,631	0.650	\$1,107,676

Tariff & element of service	Pricing Unit	2012 kWh/ cust no/ kVA	Proposed Charges 2013/14	Notional revenue 2013/14
122 HV TOU Demand Network – Customer HV and LV				
Network access	\$/day/connection point	0	17.00	\$0
Maximum demand (in billing period)	c/kVA/day	0	13.915	\$0
Capacity (maximum demand in last year)	c/kVA/day	0	13.915	\$0
Energy at business times	c/kWh	0	3.303	\$0
Energy at evening times	c/kWh	0	1.791	\$0
Energy at off-peak times	c/kWh	0	0.650	\$0
Total				\$187,286,003
Total Customers		164,828		
Total Energy Consumption		2,909,291,502		

3.2.1 Side constraints

Table 3.4 sets out for each tariff class related to standard control services the expected weighted average DUOS revenue²³ for the regulatory year and the current year, as required by clause 6.18.2(b)(4) of the transitional *Rules*.

Clause 6.18.6 of the transitional *Rules* applies a side constraint on the expected weighted average revenue to be raised from tariff classes. The side constraint is given by the formula:

$$\frac{\sum_{k=1}^m d_k^t \times q_k^{ct-1}}{\sum_{k=1}^m d_k^{t-1} \times q_k^{ct-1}} \leq (1 + \Delta CPI_t) \times (1 - X_t) \times (1 + 2\%) \pm (pass\ through_t) \quad k = 1, \dots, m$$

where:

The tariff class has up to 'm' components:

- d_k^t is the proposed price for component 'k' for regulatory year 't' (e.g. 2009–10);
- d_k^{t-1} is the current price charged by the DNSP for component 'k' in regulatory year 't-1' (e.g. 2008–09);
- q_k^{ct-1} is the audited/ verifiable quantity of component 'k' of the tariff that was charged by the DNSP in calendar year 't-1' (e.g. 2008);
- X_t are the amounts as determined by the AER in table 17.8 of the final decision. If $X > 0$, then X will be set equal to zero for the purposes of the side constraint formula.
- ' $pass\ through_t$ ' represent approved pass through amounts (expressed in percentage form) with respect to regulatory year 't' as determined by the AER under rule 6.6 of the transitional chapter 6 rules and chapter 16 of this final decision.
- ΔCPI_t means the number derived, with respect to regulatory year 't', described in section 3.1.1 above.

For 2013/14, the CPI is 1.76 per cent and the X factor is -5.41 per cent. Also, the approved pass-through costs are equivalent to 0.68 per cent of the revenue cap in 2013/14, compared to -2.22 per cent in 2012/13, a difference of 2.90 per cent. Therefore, the side constraint is

²³ This is the total DUOS revenue for each class divided by the total energy delivered to each class.

12.31 per cent ($1.0176 \times 1.0541 \times 1.02 + 0.0290 - 1$). As shown in Table 3-4, the proposed average price increase for each of the three tariff classes is within the side constraint.

Table 3-4 Weighted average DUOS revenue by tariff class

Tariff class	2012/13 (c/kWh)	2013/14 (c/kWh)	Change %
Residential	5.41	6.05	11.9%
Commercial Low Voltage	6.85	7.40	8.0%
High Voltage	3.86	4.04	4.6%
Average	5.89	6.43	9.2%

3.3 Miscellaneous services charges

As set out in the AER's guideline for standard control services, prices for miscellaneous services are regulated under the standard control services revenue cap. Like the DUOS charges, the charges for miscellaneous services for year "t" are checked for compliance with the revenue constraint for year "t" by multiplying the prices for the coming year by actual sales in the previous calendar year, for each service.²⁴

Proposed charges for miscellaneous services are shown in Table 3.5. These charges, when applied to units sold in 2012, would have recovered \$1,292,507.

Table 3-5 Miscellaneous fees and charges 2013/14

Code	Description	Pricing Unit	Units 2012	Prices 2013/14	Revenue with 2013/14 prices
For visit to re-energise or de-energise a site					
501	Business hours – re-energise a site	per visit	4,585	\$54.55	\$250,120
502	After hours – re-energise a site	per visit	1,145	\$117.27	\$134,315
503	Business hours – de-energise a site	per visit	2,894	\$48.18	\$139,414
505	De-energise a site for non-payment	per visit	374	\$90.91	\$34,009
508	Field visit only (De-energise a site for non-payment)	per visit	35	\$65.45	\$2,281
Temporary connections					
520	Overhead	per installation	15	\$387.27	\$5,681
521	Standard underground	per installation	-	\$563.64	\$0
522	Free-standing underground	per installation	2	\$683.64	\$1,367
Services					
530	Overhead: remove, reposition or disconnect service	per site visit	400	\$280.00	\$112,030
531	Underground: remove, reposition or disconnect	per site visit	623	\$671.82	\$418,260

²⁴ AER 2008, *Guideline on control mechanisms for direct control services for the ACT and NSW 2009 distribution determinations*, February, p. 5.

Code	Description	Pricing Unit	Units 2012	Prices 2013/14	Revenue with 2013/14 prices
	service				
	Upgrade service from single to three phase at customer's request where load does not justify three phase				
532	Overhead	per installation	-	\$360.91	\$0
533	Underground – service cable replacement not required	per installation	9	\$360.91	\$3,359
534	Underground – service cable replacement required	per installation	5	\$671.82	\$3,359
540	Rescheduled visit (applied where a revisit to a site is necessitated by obstructed access, non-compliance with the Service and Installation Rules or the client is not ready for the scheduled work.)	per visit	-	\$185.45	\$0
550	Issue of copies of electrical drawings	per sheet or electronic equivalent	152	\$14.45	\$2,190
560	De-energising wires (to allow safe approach, for example, for tree pruning, plant operation, oversize loads, construction activities)	per occurrence	243	\$449.09	\$109,235
570	Embedded generation - charges for small embedded generators (excluding renewable < 30KW)				
	for operation and maintenance of connection assets	per annum applied to the value of connection assets appreciated each year by CPI		2.00%	\$0
	for operation and maintenance of shared network assets (if applicable)	per annum applied to the value of shared assets allocated to the generator appreciated by CPI		2.00%	\$0
Total standard control miscellaneous services					\$1,215,623

3.4 Transmission use of system charges

The AER separately regulates transmission use of system (TUOS) charges. TUOS charges are not part of ActewAGL Distribution's regulated revenue requirement. Clause 6.18.7(a) of the transitional *Rules* allows ActewAGL Distribution to pass on to customers the charges to be incurred by ActewAGL Distribution for TUOS services. Clause 6.18.7(b) of the transitional *Rules* says that the amount to be passed on must not exceed the estimated amount of the TUOS charges for the relevant regulatory year adjusted for under or over recovery in the previous regulatory year. Clause 6.18.7(c) describes the method to be applied in determining the extent of under or over recovery.

To demonstrate compliance with clause 6.18.7 of the transitional *Rules* and the AER's final decision, the AER requires ActewAGL Distribution to maintain a transmission use of system (TUOS) overs and unders account. Clause 6.18.2(b)(7) requires ActewAGL Distribution to provide information on this account as part of the annual pricing proposal. The AER requires ActewAGL to provide details of its calculations in the format set out in Table E.1 of the Final Distribution Determination. **Table 3-6** provides the required information.

Table 3-6 TUOS overs and unders account (\$'000)

	2011/12 Actual	2012/13 Estimate	2013/14 Forecast
Revenue from TUOS charges	42,136	46,016	39,184
Transmission charges to be paid to TNSPs	39,867	44,278	43,046
Avoided TUOS payments	0	0	0
Inter-DNSP payments	0	0	0
Total transmission related payments	39,867	44,278	43,046
Over (under) recovery for the financial year	2,269	1,738	-3,862
Overs and unders account			
Annual rate of interest applicable to balances	8.79%	8.79%	8.79%
Semi-annual interest rate	4.30%	4.30%	4.30%
Opening Balance	-579	1,737	3,702
Interest on opening balance	-51	153	325
Over/under recovery for financial year	2,269	1,738	-3,862
Interest on over/under recovery	98	75	-166
Closing balance	1,737	3,702	0

The forecast revenue requirement from TUOS charges for 2013/14 shown in Table 3-6 is \$39,184,284, a decrease of 14.8 per cent compared to forecast TUOS revenue for 2012/13. This amount has been calculated on the basis of TransGrid's forecast prices for 2013/14 and the forecast energy imported into the ACT.

Based on TransGrid's prices, ActewAGL Distribution has forecast TUOS costs would be \$43,046,039 in 2013/14, a decrease of 2.8 per cent. ActewAGL Distribution recovers TransGrid's TUOS charges from ACT consumers according to the energy they consume and, where possible, according to maximum demand in a month and over the year (where capacity charges apply). The cost allocations take into account the load profile of each customer class. The consumption profile used to calculate TUOS prices is the same 2012 consumption profile used to calculate DUOS prices. Also, they are adjusted for the over or under recovery of TUOS charges in the previous regulatory years.

As the energy transported in 2013/14 is forecast to be about 0.7 per cent lower²⁵ than the actual energy transported in the 2012 calendar year, the amount of TUOS to be recovered in prices using the load in 2012 has been adjusted upward to take account of that difference.²⁶ The TUOS prices would have recovered revenue of \$39,460,285 under the 2012 profile as shown in Table 3.7.

²⁵ This uses the forecast in the AER's Final Decision.

²⁶ AER, *Final decision, Control mechanisms for direct control services for the ACT and NSW 2009 distribution determinations*, February 2008, part 5.8.

Table 3-7 Transmission use of system charges 2013/14

Description	Pricing Unit	2012 kWh/ cust no /kVA	Price 2013/14	Notional TUOS revenue
Residential tariffs				
010 Residential Basic Network				
Network access charge	cents/day	130,167	0.00	\$0
Energy consumption	cents/kWh	923,491,037	1.421	\$13,122,808
015 Residential TOU Network				
Network access charge	cents/day	7,325	0.00	\$0
Energy at max times	cents/kWh	10,050,658	0.196	\$19,699
Energy at mid times	cents/kWh	13,884,998	0.149	\$20,689
Energy at economy times	cents/kWh	10,761,240	0.118	\$12,698
020 Residential 5000 Network				
Network access charge	cents/day	8,225	0.00	\$0
Energy for the first 60 kWh per day	cents/kWh	45,724,339	1.421	\$649,743
Energy above 60 kWh per day	cents/kWh	1,156,844	1.421	\$16,439
030 Residential with Heat Pump Network				
Network access charge	cents/day	4,536	0.00	\$0
Energy for the first 165 kWh per day	cents/kWh	80,019,029	1.177	\$941,824
Energy above 165 kWh per day	cents/kWh	1,168,768	1.421	\$16,608
060 Off-Peak (1) Night Network				
Energy consumption	cents/kWh	14,884,138	1.335	\$198,703
070 Off-Peak (3) Day & Night Network				
Energy consumption	cents/kWh	91,278,494	1.736	\$1,584,595
Renewable Energy Generation				
Energy consumption (Obsolete)	cents/kWh	3,208,260	0.000	\$0
Energy exports	cents/kWh		-0.50	-0.50
Commercial low voltage tariffs				
040 General Network				
Network access charge	cents/day	11,987	0.00	\$0
Energy for the first 330 kWh per day	cents/kWh	263,694,449	1.783	\$4,701,672
Energy above 330 kWh per day	cents/kWh	25,852,797	1.682	\$434,844
135 Small Unmetered Loads Network				
Network access charge	cents/day	23	0.00	\$0
Energy consumption	cents/kWh	1,513,183	1.811	\$27,404
141 Internal Network (Discontinued- move to General Network)				
Network access charge	cents/day	96	0.00	\$0
Energy consumption	cents/kWh	798,363	1.783	\$14,235
080 Streetlighting Network				
Network access charge	cents/day	17	0.00	\$0
Energy consumption	cents/kWh	41,722,136	1.084	\$452,268
090 General TOU Network				
Network access charge	cents/day	662	0.00	\$0
Energy at business times	cents/kWh	25,775,332	0.426	\$109,803
Energy at evening times	cents/kWh	11,684,169	0.248	\$28,977
Energy at off-peak times	cents/kWh	29,790,219	0.172	\$51,239
Low voltage time of use demand network				
101 LV TOU kVA Demand Network				
Network access per connection point	cents/day	1,741	0.00	\$0
Maximum demand charge	c/kVA/day	233,561	6.079	\$5,196,542
Energy at business times	cents/kWh	372,478,364	0.817	\$3,043,148
Energy at evening times	cents/kWh	132,328,674	0.782	\$1,034,810
Energy at off-peak times	cents/kWh	404,991,258	0.725	\$2,936,187

Description	Pricing Unit	2012 kWh/ cust no /kVA	Price 2013/14	Notional TUOS revenue
103 LV TOU Capacity Network				
Network access per connection point	cents/day	25	0.00	\$0
Maximum demand charge	c/kVA/day	7,227	2.918	\$77,182
Capacity charge	c/kVA/day	8,324	2.918	\$88,902
Energy at business times	cents/kWh	13,606,712	0.817	\$111,167
Energy at evening times	cents/kWh	5,502,140	0.782	\$43,027
Energy at off-peak times	cents/kWh	18,858,594	0.627	\$118,243
High voltage tariffs				
High voltage time of use demand network with ActewAGL low voltage network				
111 HV TOU Demand Network				
Network access per connection point	\$/day	1	0.00	\$0
Maximum demand charge	c/kVA/day	1,605	2.885	\$16,950
Capacity charge	c/kVA/day	1,923	3.308	\$23,288
Energy at business times	cents/kWh	2,833,657	0.937	\$26,551
Energy at evening times	cents/kWh	1,163,428	0.779	\$9,063
Energy at off-peak times	cents/kWh	3,493,430	0.720	\$25,153
112 HV TOU Demand Network – Customer HV				
Network access per connection point	\$/day	0	0.00	\$0
Maximum demand charge	c/kVA/day	0	2.885	\$0
Capacity charge	c/kVA/day	0	3.308	\$0
Energy at business times	cents/kWh	0	0.937	\$0
Energy at evening times	cents/kWh	0	0.779	\$0
Energy at off-peak times	cents/kWh	0	0.720	\$0
High voltage time of use demand network without ActewAGL low voltage network				
121 HV TOU Demand Network – Customer LV				
Network access per connection point	\$/day	22	0.00	\$0
Maximum demand charge	c/kVA/day	67,781	2.885	\$715,704
Capacity charge	c/kVA/day	80,853	2.885	\$853,731
Energy at business times	cents/kWh	135,284,203	0.817	\$1,105,272
Energy at evening times	cents/kWh	51,880,958	0.779	\$404,153
Energy at off-peak times	cents/kWh	170,411,631	0.720	\$1,226,964
122 HV TOU Demand Network – Customer HV and LV				
Network access per connection point	\$/day	0	0.00	\$0
Maximum demand charge	c/kVA/day	0	2.885	\$0
Capacity charge	c/kVA/day	0	2.885	\$0
Energy at business times	cents/kWh	0	0.817	\$0
Energy at evening times	cents/kWh	0	0.779	\$0
Energy at off-peak times	cents/kWh	0	0.720	\$0
Total				\$39,460,285

3.5 Network use of system charges

Network use of system (NUOS) charges for 2013/14 comprise the distribution use of system (DUOS) charges and transmission use of system (TUOS) charges. The proposed NUIS charges are shown in Table 3-8. All charges exclude GST.

Table 3-8 Network use of system charges 2013/14

Code	Description	Pricing Unit	Distribution Charges 2013/14	Transmission Charges 2013/14	Network Charges 2013/14
Residential tariffs					
010	Residential Basic Network				
	Network access charge	cents/day	21.44	0.00	21.44
	Energy consumption	cents/kWh	5.649	1.421	7.070
015	Residential TOU Network				
	Network access charge	cents/day	21.44	0.00	21.44
	Energy consumption at max times	cents/kWh	9.894	0.196	10.090
	Energy consumption at mid times	cents/kWh	5.511	0.149	5.660
	Energy consumption at economy times	cents/kWh	4.042	0.118	4.160
020	Residential 5000 Network				
	Network access charge	cents/day	42.64	0.00	42.64
	Energy consumption for the first 60 kWh per day	cents/kWh	4.109	1.421	5.530
	Energy consumption above 60 kWh per day	cents/kWh	5.649	1.421	7.070
030	Residential with Heat Pump Network				
	Network access charge	cents/day	85.34	0.00	85.34
	Energy consumption for the first 165 kWh per day	cents/kWh	2.893	1.177	4.070
	Energy consumption above 165 kWh per day	cents/kWh	5.649	1.421	7.070
060	Off-Peak (1) Night Network				
	Energy consumption	cents/kWh	0.085	1.335	1.420
070	Off-Peak (3) Day & Night Network				
	Energy consumption	cents/kWh	0.204	1.736	1.940
	Renewable Energy Generation				
	Energy consumption (Obsolete)	cents/kWh	0.000	0.000	0.000
	Energy exports	cents/kWh	0.000	-0.50	-0.50
Commercial low voltage tariffs					
040	General Network				
	Network access charge	cents/day	42.33	0.00	42.33
	Energy consumption for the first 330 kWh per day	cents/kWh	9.097	1.783	10.880
	Energy consumption above 330 kWh per day	cents/kWh	12.818	1.682	14.500
135	Small Unmetered Loads Network				
	Network access charge	cents/day	35.80	0.00	35.80
	Energy consumption	cents/kWh	10.738	1.811	12.549
141	Internal Network (Discontinued- move to General Network)				
	Network access charge	cents/day	42.33	0.00	42.33
	Energy consumption	cents/kWh	9.097	1.783	10.880
080	Streetlighting Network				
	Network access charge	cents/day	43.00	0.00	43.00
	Energy consumption	cents/kWh	6.126	1.084	7.210
090	General TOU Network				
	Network access charge	cents/day	42.33	0.00	42.33
	Energy consumption at business times	cents/kWh	17.134	0.426	17.560
	Energy consumption at evening times	cents/kWh	8.532	0.248	8.780
	Energy consumption at off-peak times	cents/kWh	3.478	0.172	3.650
Low voltage time of use demand network					
101	LV TOU kVA Demand Network				
	Network access charge per connection point	cents/day	49.00	0.00	49.00

Code	Description	Pricing Unit	Distribution Charges 2013/14	Transmission Charges 2013/14	Network Charges 2013/14
	Maximum demand charge	c/kVA/day	39.021	6.079	45.100
	Energy consumption at business times	cents/kWh	4.733	0.817	5.550
	Energy consumption at evening times	cents/kWh	2.798	0.782	3.580
	Energy consumption at off-peak times	cents/kWh	1.045	0.725	1.770
103	LV TOU Capacity Network				
	Network access charge per connection point	cents/day	49.00	0.00	49.00
	Maximum demand charge	c/kVA/day	18.082	2.918	21.000
	Capacity charge	c/kVA/day	18.082	2.918	21.000
	Energy consumption at business times	cents/kWh	4.733	0.817	5.550
	Energy consumption at evening times	cents/kWh	2.798	0.782	3.580
	Energy consumption at off-peak times	cents/kWh	1.143	0.627	1.770
High voltage tariffs					
High voltage time of use demand network with ActewAGL low voltage network					
111	HV TOU Demand Network				
	Network access charge per connection point	\$/day	17.00	0.00	17.00
	Maximum demand charge	c/kVA/day	14.915	2.885	17.800
	Capacity charge	c/kVA/day	14.492	3.308	17.800
	Energy consumption at business times	cents/kWh	3.583	0.937	4.520
	Energy consumption at evening times	cents/kWh	2.141	0.779	2.920
	Energy consumption at off-peak times	cents/kWh	0.790	0.720	1.510
112	HV TOU Demand Network – Customer HV				
	Network access charge per connection point	\$/day	17.00	0.00	17.00
	Maximum demand charge	c/kVA/day	13.915	2.885	16.800
	Capacity charge	c/kVA/day	13.492	3.308	16.800
	Energy consumption at business times	cents/kWh	3.583	0.937	4.520
	Energy consumption at evening times	cents/kWh	2.141	0.779	2.920
	Energy consumption at off-peak times	cents/kWh	0.790	0.720	1.510
High voltage time of use demand network without ActewAGL low voltage network					
121	HV TOU Demand Network – Customer LV				
	Network access charge per connection point	\$/day	17.00	0.00	17.00
	Maximum demand charge	c/kVA/day	14.915	2.885	17.800
	Capacity charge	c/kVA/day	14.915	2.885	17.800
	Energy consumption at business times	cents/kWh	3.303	0.817	4.120
	Energy consumption at evening times	cents/kWh	1.791	0.779	2.570
	Energy consumption at off-peak times	cents/kWh	0.650	0.720	1.370
122	HV TOU Demand Network – Customer HV and LV				
	Network access charge per connection point	\$/day	17.00	0.00	17.00
	Maximum demand charge	c/kVA/day	13.915	2.885	16.800
	Capacity charge	c/kVA/day	13.915	2.885	16.800
	Energy consumption at business times	cents/kWh	3.303	0.817	4.120
	Energy consumption at evening times	cents/kWh	1.791	0.779	2.570
	Energy consumption at off-peak times	cents/kWh	0.650	0.720	1.370

3.6 Changes to network tariffs

The transitional *Rules* (6.18.2(b)(8)) require an explanation of the nature and extent of changes from the previous regulatory year. Table 3.9 compares the network charges in

2013/14 with those in 2012/13. The average change in network charges is shown in cents per kWh and as a percentage for an average consumer for each tariff.²⁷

ActewAGL Distribution proposes to change the buy-back arrangements for consumers installing small embedded PV generation systems (less than 30KW). The current arrangements specified in ActewAGL Distribution's schedule of charges is as follows:

If a customer has a grid-connected renewable energy generator and the customer is not receiving the ACT feed-in tariff, the following arrangements shall apply:

The customer shall pay the published network charge for the gross amount of energy imported from ActewAGL Distribution's network.

ActewAGL Distribution will pay to the customer's retailer an amount equal to the customer's network energy tariff rate on the gross amount of energy exported.

If the customer is on a time-of-use tariff, the customer's network energy tariff rate for the purpose of paying the customer's retailer shall be the middle rate (mid rate for residential customers and evening rate for commercial customers).

If the customer is on a tariff with two or more price steps, the customer's network energy tariff rate for the purpose of paying the customer's retailer shall be the first rate applied.

The customer shall continue to pay the network access charge.

This arrangement is available only to customers with less than 30 kilowatts installed capacity of renewable generation.

In all other circumstances where a customer has a grid-connected renewable energy generator with an installed capacity of less than 30 kilowatts, including where the customer is receiving the ACT feed-in tariff, the following arrangements shall apply:

The customer shall pay the published network charge for the gross amount of energy imported from ActewAGL Distribution's network.

ActewAGL Distribution shall not charge the customer for the use of the network for the energy exported.

The customer shall continue to pay the network access charge.

It is proposed to move to net metering and pay the consumer the avoided TUOS costs for energy exported into the electricity network. To implement the change, ActewAGL Distribution would change the relevant section of its schedule of charges to read:

If a customer has a grid-connected renewable energy generator with a net metering facility and the customer is not receiving the ACT feed-in tariff, the following arrangements shall apply to PV installations:

The customer shall pay the published network charge for energy imported from ActewAGL Distribution's network (based upon the customer's meter reading).

ActewAGL Distribution will pay to the customer's retailer an amount equal to ActewAGL Distribution's estimated avoided cost of TUOS charges on energy exported into the electricity network (based upon the customer's meter reading).

The customer shall continue to pay the network access charge.

²⁷ The average change in network charges is calculated by determining the average revenue for each tariff using the prices for each year and taking the difference. The percentage change is this difference divided by the average revenue with 2012/13 prices. Where there are no customers, it is not possible to calculate the average price.

This arrangement is available only to customers with less than 30 kilowatts installed capacity of renewable generation with a net metering facility able to record energy imported and exported into the electricity network.

Customers with a grid-connected renewable energy generator which was connected on or before 30 June 2013 may continue with the existing arrangements applicable to that customer.

In all other circumstances where a customer has a grid-connected renewable energy generator with an installed capacity of less than 30 kilowatts, including where the customer is receiving the ACT feed-in tariff, the following arrangements shall apply:

The customer shall pay the published network charge for the gross amount of energy imported from ActewAGL Distribution's network.

ActewAGL Distribution shall not charge the customer for the use of the network for the energy exported.

The customer shall continue to pay the network access charge.

"Energy exported" means energy generated by a photovoltaic system that results in energy flowing from the customer's premises into the electricity network.

Table 3-9 Changes in network charges

Code	Description	Pricing Unit	Network Charges 2012/13	Network Charges 2013/14	Average change c/kWh	Average change %
Residential tariffs						
010	Residential Basic Network				0.467	6.1%
	Network access charge	cents/day	16.45	21.44		
	Energy consumption	cents/kWh	6.86	7.07		
015	Residential TOU Network				0.620	8.2%
	Network access charge	cents/day	16.45	21.44		
	Energy consumption at max times	cents/kWh	9.55	10.09		
	Energy consumption at mid times	cents/kWh	5.52	5.66		
	Energy consumption at economy times	cents/kWh	4.09	4.16		
020	Residential 5000 Network				0.530	6.8%
	Network access charge	cents/day	37.65	42.64		
	Energy consumption for the first 60 kWh per day	cents/kWh	5.32	5.53		
	Energy consumption above 60 kWh per day	cents/kWh	6.86	7.07		
030	Residential with Heat Pump Network				0.312	5.6%
	Network access charge	cents/day	80.35	85.34		
	Energy consumption for the first 165 kWh per day	cents/kWh	3.86	4.07		
	Energy consumption above 165 kWh per day	cents/kWh	6.86	7.07		
060	Off-Peak (1) Night Network				-0.120	-7.8%
	Energy consumption	cents/kWh	1.54	1.42		
070	Off-Peak (3) Day & Night Network				-0.120	-5.8%
	Energy consumption	cents/kWh	2.06	1.94		
	Renewable Energy Generation					
	Energy consumption (Obsolete)	cents/kWh	0.00	0.00		
	Energy exports	cents/kWh	0.00	-0.50		
Commercial low voltage						
040	General Network				0.634	5.7%
	Network access charge	cents/day	32.71	42.33		
	Energy consumption for the first 330 kWh per day	cents/kWh	10.42	10.88		
	Energy consumption above 330 kWh per day	cents/kWh	13.72	14.50		
135	Small Unmetered Loads Network				0.479	3.9%
	Network access charge	cents/day	26.80	35.80		
	Energy consumption	cents/kWh	12.12	12.549		
141	Internal Network (Discontinued – move to General Network)				0.883	7.4%
	Network access charge	cents/day	32.71	42.33		
	Energy consumption	cents/kWh	10.42	10.88		
080	Streetlighting Network				0.202	2.9%
	Network access charge	cents/day	33.00	43.00		
	Energy consumption	cents/kWh	7.01	7.21		
090	General TOU Network				0.473	4.9%
	Network access charge	cents/day	32.71	42.33		
	Energy consumption at business times	cents/kWh	16.56	17.56		
	Energy consumption at evening times	cents/kWh	8.44	8.78		
	Energy consumption at off-peak times	cents/kWh	3.66	3.65		

Code	Description	Pricing Unit	Network Charges 2012/13	Network Charges 2013/14	Average change c/kWh	Average change %
Low voltage time of use demand network						
101	LV TOU kVA Demand Network				0.176	2.3%
	Network access charge per connection point	cents/day	39.00	49.00		
	Maximum demand charge	c/kVA/day	42.80	45.10		
	Energy consumption at business times	cents/kWh	5.46	5.55		
	Energy consumption at evening times	cents/kWh	3.70	3.58		
	Energy consumption at off-peak times	cents/kWh	1.92	1.77		
103	LV TOU Capacity Network				0.138	2.1%
	Network access charge per connection point	cents/day	39.00	49.00		
	Maximum demand charge	c/kVA/day	19.70	21.00		
	Capacity charge	c/kVA/day	19.70	21.00		
	Energy consumption at business times	cents/kWh	5.46	5.55		
	Energy consumption at evening times	cents/kWh	3.70	3.58		
	Energy consumption at off-peak times	cents/kWh	1.92	1.77		
High voltage tariffs						
High voltage time of use demand network with ActewAGL low voltage network						
111	HV TOU Demand Network				0.108	1.8%
	Network access charge per connection point	\$/day	14.00	17.00		
	Maximum demand charge	c/kVA/day	16.80	17.80		
	Capacity charge	c/kVA/day	16.80	17.80		
	Energy consumption at business times	cents/kWh	4.54	4.52		
	Energy consumption at evening times	cents/kWh	3.02	2.92		
	Energy consumption at off-peak times	cents/kWh	1.63	1.51		
112	HV TOU Demand Network – Customer HV					
	Network access charge per connection point	\$/day	14.00	17.00		
	Maximum demand charge	c/kVA/day	15.80	16.80		
	Capacity charge	c/kVA/day	15.80	16.80		
	Energy consumption at business times	cents/kWh	4.54	4.52		
	Energy consumption at evening times	cents/kWh	3.02	2.92		
	Energy consumption at off-peak times	cents/kWh	1.63	1.51		
High voltage time of use demand network without ActewAGL low voltage network						
121	HV TOU Demand Network – Customer LV				0.080	1.5%
	Network access charge per connection point	\$/day	14.00	17.00		
	Maximum demand charge	c/kVA/day	16.80	17.80		
	Capacity charge	c/kVA/day	16.80	17.80		
	Energy consumption at business times	cents/kWh	4.14	4.12		
	Energy consumption at evening times	cents/kWh	2.67	2.57		
	Energy consumption at off-peak times	cents/kWh	1.49	1.37		
122	HV TOU Demand Network – Customer HV and LV					
	Network access charge per connection point	\$/day	14.00	17.00		
	Maximum demand charge	c/kVA/day	15.80	16.80		
	Capacity charge	c/kVA/day	15.80	16.80		
	Energy consumption at business times	cents/kWh	4.14	4.12		
	Energy consumption at evening times	cents/kWh	2.67	2.57		
	Energy consumption at off-peak times	cents/kWh	1.49	1.37		

The increases shown in the Table 3-9 reflect the increases in both DUOS and TUOS charges for 2013/14. The CPI of 1.76 per cent and the X factor of -5.41 per cent raise the MAAR by 7.3 per cent. Offsetting this, there is a 14.9 per cent reduction in average TUOS charges in 2013/14. The shift from a negative pass through in 2012/13 of 2.22 per cent to a positive pass through in 2013/14 of 0.68 per cent has increased average DUOS charges by 2.9 per cent.

Table 3.10 compares the miscellaneous fees and charges for 2013/14 with the comparable charges for 2012/13. The average increase in these miscellaneous fees and charges is around 3.9 per cent.

Table 3-10 Changes in miscellaneous services charges

Code	Description	Pricing Unit	Prices 2012/13	Prices 2013/14	Change %
For visit to re-energise or de-energise a site					
501	Business hours – re-energise a site	per visit	\$52.73	\$54.55	3.5%
502	After hours – re-energise a site	per visit	\$112.73	\$117.27	4.0%
503	Business hours – de-energise a site	per visit	\$46.36	\$48.18	3.9%
505	De-energise a site for non-payment	per visit	\$87.27	\$90.91	4.2%
508	Field visit only (De-energise a site for non-payment)	per visit	\$62.73	\$65.45	4.3%
Temporary connections					
520	Overhead	per installation	\$372.73	\$387.27	3.9%
521	Standard underground	per installation	\$541.82	\$563.64	4.0%
522	Free-standing underground	per installation	\$657.27	\$683.64	4.0%
Services					
530	Overhead: remove, reposition or disconnect service	per site visit	\$269.09	\$280.00	4.1%
531	Underground: remove, reposition or disconnect service	per site visit	\$646.36	\$671.82	3.9%
Upgrade service from single to three phase at customer's request where load does not justify three phase					
532	Overhead	per installation	\$347.27	\$360.91	3.9%
533	Underground – service cable replacement not required	per installation	\$347.27	\$360.91	3.9%
534	Underground – service cable replacement required	per installation	\$646.36	\$671.82	3.9%
540	Rescheduled visit (applied where a revisit to a site is necessitated by obstructed access, non-compliance with the Service and Installation Rules or the client is not ready for the scheduled work.)		\$178.18	\$185.45	4.1%
550	Issue of copies of electrical drawings	per sheet or electronic equivalent	\$13.91	\$14.45	3.9%
560	De-energising wires (to allow safe approach, for example, for tree pruning, plant operation, oversize loads, construction activities)	per occurrence	\$431.82	\$449.09	4.0%
570	Embedded generation - operational and maintenance charges for small embedded generators (other renewable <30KW)				
	for connection assets		2.0%	2.0%	0.0%
	for shared network assets (if applicable)		2.0%	2.0%	0.0%

4 Charges for metering services

4.1 The structure and basis of ActewAGL Distribution's metering charges

ActewAGL Distribution's schedule of metering charges comprises daily charges for the provision of each of four different types of manually read metering services and charges for special meter reads. Meters for residential consumers are usually read quarterly while those for small low voltage commercial consumers (consuming less than 160 MWh pa) are usually read monthly. Metering services for larger low voltage consumers and for high voltage consumers are open to competition and are not regulated.

The metering charges for small consumers are set on a cost reflective basis. The costs of meter reading and meters (including depreciation, return on capital, installation and maintenance and administration) have been calculated for each meter type. The costs are applied to the number of meters of each type to derive the total cost for each meter type. The meter types are each allocated to a class of metering service and the total cost of each class is allocated to the customers in that class to estimate the average cost of the metering service to each class.

4.2 Metering services charges for 2013/14

Prices for ActewAGL Distribution's alternative control services are regulated using a maximum revenue cap.²⁸ This cap is applied to metering revenue that would have been earned from the prices for the coming financial year if they had been applied to the quantities sold in the services provided in the previous calendar year.

Table 4-1 shows the X factor for 2013/14 and CPI factors applied to calculate the revenue cap for alternative control services. The approach is similar to that applied in Table 3-1 to calculate the MAAR.

²⁸ AER 2009, *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision*, April, p. 146

Table 4-1 Alternative control services revenue cap

	<i>Alternative services revenue in previous year</i>	<i>X Factor</i>	<i>Sum of CPI indices</i>	<i>CPI</i>	<i>Alternative services revenue cap</i>
2008/09			631.8		\$5,666,539
2009/10	\$5,666,539	-29.30%	659.3	4.35%	\$7,645,746
2010/11	\$7,645,746	0.00%	671.3	1.82%	\$7,784,907
2011/12	\$7,784,907	0.00%	690.4	2.85%	\$8,006,405
2012/13	\$8,006,405	0.00%	713.8	3.39%	\$8,277,770
2012/13	New CPI base		397.1		
2013/14	\$8,277,770	0.00%	404.1	1.76%	\$8,423,688

As shown in Table 4-1, the alternative control services revenue cap for 2013/14 is \$8,423,688. To demonstrate compliance with this revenue cap, ActewAGL Distribution has prepared the required schedule of metering charges showing the revenue from each charge if that charge had applied in to the 2012 calendar year quantities.²⁹ The schedule shows that the revenue from these charges does not exceed the revenue cap for alternative control services. The metering charges and the forecast revenue from those charges are shown in Table 4-2.

²⁹ AER 2009, *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision*, April, p. 153.

Table 4-2 Metering charges 2013/14

Code	Description	Pricing Unit	Units 2012	Prices excl. GST 2013/14	Revenue
MP1	Quarterly basic metering rate Accumulation and time-of-use meters read quarterly	cents per day per NMI	150,242	12.96	\$7,126,514
MP2	Monthly basic metering rate Accumulation and time-of-use meters read monthly	cents per day per NMI	11,712	22.67	\$971,734
MP3	Time-of-use metering rate Time-of-use meters read monthly	cents per day per NMI	460	22.67	\$38,128
MP4	Monthly manually-read interval metering rate Interval meters recording at either 15 or 30 minute intervals, read manually and processed monthly	\$ per day per NMI	266	1.83	\$177,993
MP5	Internal metering rate Sites entitled to the Internal Network charge	cents per day per NMI	95	22.67	\$7,872
MP6	Quarterly manually-read interval metering rate Interval meters recording at either 15 or 30 minute intervals, read manually and processed quarterly	cents per day per NMI	0	52.20	\$0
504	Meter test during business hours (refunded if the meter proves to be faulty)	per test	18	67.27	\$1,223
506	Special meter reading , or check read deposit (refunded if the original reading was incorrect).	per read	0	34.55	\$0
507	Install interval meter at customer's request	per meter	16	64.64	\$1,034
509	Replace meter to facilitate renewable energy installation	per meter	1,533	64.64	\$99,087
Total			164,341		\$8,423,585

*National Meter Identifier.

The revenue from the 2012 units of \$8,423,585 is less than the revenue cap for alternative control services of \$8,423,688. Therefore, the charges comply with the AER's Final Distribution Determination on alternative control services.

4.3 Changes in metering charges

There are no changes to the metering services offered in 2013/14. Table 4-3 compares the metering charges for 2013/14 with those for 2012/13.

Table 4-3 Changes in metering charges

Code	Description	Unit	Prices 2012/13	Prices 2013/14	Change %
MP1	Quarterly basic metering rate Accumulation and time-of-use meters read quarterly	cents per day per NMI	12.75	12.96	1.6%
MP2	Monthly basic metering rate Accumulation and time-of-use meters read monthly	cents per day per NMI	22.29	22.67	1.7%
MP3	Time-of-use metering rate Time-of-use meters read monthly	cents per day per NMI	22.29	22.67	1.7%
MP4	Monthly manually-read interval metering rate Interval meters recording at either 15- or 30-minute intervals, read manually and processed monthly	\$ per day per NMI	\$1.81	1.83	1.1%
MP5	Internal metering rate Sites entitled to the Internal Network charge	cents per day per NMI	22.29	22.67	1.7%
MP6	Quarterly manually-read interval metering rate Interval meters recording at either 15- or 30-minute intervals, read manually and processed quarterly	cents per day per NMI	52.00	52.20	0.4%
504	Meter test during business hours (refunded if the meter proves to be faulty)	Per test	\$65.45	67.27	2.8%
506	Special meter reading , or check read deposit (refunded if the original reading was incorrect).	per read	\$33.64	34.55	2.7%
507	Install interval meter at customer's request	per meter	\$63.00	64.64	2.6%
509	Replace meter to facilitate renewable energy installation	per meter	\$63.00	64.64	2.6%

5 Indicative customer impacts

5.1 Changes in network and metering charges

ActewAGL Distribution's customers who consume less than 160 MWh per annum are most likely to be subject to charges for both standard control services (network tariffs) and alternative control services (regulated metering service charges). Table 5-1 shows network charges (standard control services and TUOS) plus metering service charges (alternative control services) for 2013/14 and the comparable charges for 2012/13, excluding GST. High voltage charges have been excluded as ActewAGL Distribution does not provide metering services to high voltage customers. Metering services to customers consuming more than 160 MWh per annum are open to competition and not regulated.

Standard control services and alternative control services are shown separately in ActewAGL Distribution's pricing schedule and are charged separately to retailers.

Table 5-1 Network and metering charges 2013/14

Code	Description	Unit	Network & metering charges 2012/13	Network & metering charges 2013/14	Average change c/kWh	Average change %
Residential tariffs						
010	Residential Basic Network				0.48	5.7%
	Network access charge	cents/day	29.20	34.40		
	Energy consumption	cents/kWh	6.86	7.07		
015	Residential TOU Network				0.64	7.5%
	Network access charge	cents/day	29.20	34.40		
	Energy consumption at max times	cents/kWh	9.55	10.09		
	Energy consumption at mid times	cents/kWh	5.52	5.66		
	Energy consumption at economy times	cents/kWh	4.09	4.16		
020	Residential 5000 Network				0.54	6.3%
	Network access charge	cents/day	50.40	55.60		
	Energy consumption for the first 60 kWh per day	cents/kWh	5.32	5.53		
	Energy consumption above 60 kWh per day	cents/kWh	6.86	7.07		
030	Residential with Heat Pump Network				0.32	5.4%
	Network access charge	cents/day	93.10	98.30		
	Energy consumption for the first 165 kWh per day	cents/kWh	3.86	4.07		
	Energy consumption above 165 kWh per day	cents/kWh	6.86	7.07		
060	Off-Peak (1) Night Network				-0.12	-7.8%
	Energy consumption	cents/kWh	1.54	1.42		
070	Off-Peak (3) Day & Night Network				-0.12	-5.8%
	Energy consumption	cents/kWh	2.06	1.94		
	Renewable Energy Generation					
	Energy consumption (Obsolete)	cents/kWh	0.00	0.00		
	Energy exports	cents/kWh	0.00	-0.50		

Code	Description	Unit	Network & metering charges 2012/13	Network & metering charges 2013/14	Average change c/kWh	Average change %
Commercial low voltage tariffs						
040	General Network				0.64	5.5%
	Network access charge	cents/day	55.00	65.00		
	Energy consumption for the first 330 kWh per day	cents/kWh	10.42	10.88		
	Energy consumption above 330 kWh per day	cents/kWh	13.72	14.50		
135	Small Unmetered Loads Network				0.39	3.2%
	Network access charge	cents/day	26.80	35.80		
	Energy consumption	cents/kWh	12.12	12.549		
080	Streetlighting Network				0.20	2.9%
	Network access charge	cents/day	55.29	65.67		
	Energy consumption	cents/kWh	7.01	7.21		
090	General TOU Network				0.47	4.9%
	Network access charge	cents/day	55.00	65.00		
	Energy consumption at business times	cents/kWh	16.56	17.56		
	Energy consumption at evening times	cents/kWh	8.44	8.78		
	Energy consumption at off-peak times	cents/kWh	3.66	3.65		
Low voltage time of use demand network						
101	LV TOU kVA Demand Network				0.18	2.3%
	Network access charge per connection point	cents/day	220	232		
	Maximum demand charge	c/kVA/day	42.80	45.10		
	Energy consumption at business times	cents/kWh	5.46	5.55		
	Energy consumption at evening times	cents/kWh	3.70	3.58		
	Energy consumption at off-peak times	cents/kWh	1.92	1.77		
103	LV TOU Capacity Network				0.14	2.1%
	Network access charge per connection point	cents/day	220	232		
	Maximum demand charge	c/kVA/day	19.70	21.00		
	Capacity charge	c/kVA/day	19.70	21.00		
	Energy consumption at business times	cents/kWh	5.46	5.55		
	Energy consumption at evening times	cents/kWh	3.70	3.58		
	Energy consumption at off-peak times	cents/kWh	1.92	1.77		

5.2 Estimated impacts on average customer electricity bills

The proposed 2013/14 increases in network and metering charges would raise the electricity bill for an average residential customer consuming 7000 kWh on the Residential Basic network tariff by \$0.71 per week (including GST), a real increase of 3.9 per cent (5.7 per cent nominal). For a commercial customer consuming 30 MWh per annum on the General network charge, the network and metering price increases would raise their electricity bill by \$3.69 per week (including GST) or 3.4 per cent real (5.2 per cent nominal).

Attachment 1: Compliance with transitional Rules requirements

Table A1.1 provides a checklist of where the relevant requirements of Part I of the transitional *Rules* are met in this document. One of the transitional *Rules* requirements is that the pricing proposal demonstrates compliance with any applicable distribution determination (clause 6.18.2(7)). Table A1.2 provides a separate checklist of the relevant requirements from the AER's *Australian Capital Territory Distribution Determination 2009/10 to 2013/14: Final Decision* and where they are addressed in this document.

Table A1.1: Checklist of transitional Rules pricing proposal requirements

Rules	Requirement	Coverage in this document
6.18.2 (b)	The pricing proposal must: (1) set out the tariff classes that are to apply for the relevant regulatory year; and (2) set out the proposed tariffs for each tariff class; and (3) set out, for each proposed tariff, the charging parameters and the elements of service to which each charging parameter relates; and (4) set out, for each tariff class related to standard control services, the expected weighted average revenue for the relevant regulatory year and also for the current regulatory year; and (5) set out the nature of any variation or adjustment to the tariff that could occur during the course of the regulatory year and the basis on which it could occur; and (6) set out how charges incurred by the Distribution Network Service Provider for transmission use of system services are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those charges in the previous regulatory year; and (7) demonstrate compliance with the Rules and any applicable distribution determination; and (8) describe the nature and extent of change from the previous regulatory year and demonstrate that the changes comply with the Rules and any applicable distribution determination.	(1) The tariff classes for standard control services are set out in Section 2.1. (2) Distribution tariffs are in Table 3-3. Metering charges are in Table 4-2. (3) Section 2.1 and Tables 2.1, 2.2 and 2.3 set out each charging parameter and the element of service to which it relates. (4) Table 3.4 shows the weighted average DOUS revenue for 2012/13 and 2013/14. (5) ActewAGL Distribution does not propose any variations or adjustments to tariffs for 2013/14. Any variations would be made in accordance with the cost pass through mechanisms in the AER's final determination. (6) Section 3.4 provides an explanation of how TUOS charges are passed on to customers, and ActewAGL Distribution's adjustment for under recovery of TUOS costs in 2013/14. TUOS charges are provided in Table 3-7 (7) Table A1.2 below provides the checklist for relevant requirements from the 2009-14 distribution determination. (8) Sections 3.6 and 4.3 contain descriptions of the nature and extent of changes from the 2012/13 regulatory year. Table 3.3 and Table 3-5 demonstrate that revenue from 2013/14 prices matches allowable revenue calculated in section 3.1.1
6.18.3	(a) A pricing proposal must define the tariff classes into which customers for direct control services are divided. (b) Each customer for direct control services must be a member of 1 or more tariff classes. (c) Separate tariff classes must be constituted for customers to whom standard control services are supplied and customers	(a) Section 2.1 provides an explanation of the tariff classes for standard control services. Section 4.1 provides an explanation of the categories of charges for alternative control services. (b) Each customer is on one or more tariffs within one or more tariff classes for which qualify (c) Separate tariff classes and charges are specified for standard control services in

	to whom alternative control services are supplied (but a customer for both standard control services and alternative control services may be a member of 2 or more tariff classes). (d) A tariff class must be constituted with regard to: (i) the need to group customers together on an economically efficient basis; and (ii) the need to avoid unnecessary transaction costs.	Table 3-3 and Table 3-5 and alternative control services in Table 4-2. (d) Section 2.1 contains an explanation of the basis of the tariff classes.
6.18.4	(a) In formulating provisions of a distribution determination governing the assignment of customers to tariff classes or the re-assignment of customers from one tariff class to another, the AER must have regard to the following principles: (1) customers should be assigned to tariff classes on the basis of one or more of the following factors: (i) the nature and extent of their usage; (ii) the nature of their connection to the network; (iii) whether remotely-read interval metering or other similar metering technology has been installed at the customer's premises as a result of a regulatory obligation or requirement; (2) customers with a similar connection and usage profile should be treated on an equal basis; (3) however, customers with micro-generation facilities should be treated no less favourably than customers without such facilities but with a similar load profile; (4) a Distribution Network Service Provider's decision to assign a customer to a particular tariff class, or to re-assign a customer from one tariff class to another should be subject to an effective system of assessment and review.	(a) Appendix A of the AER's final decision sets out the procedures for assigning or re-assigning customers to tariff classes.
6.18.5	(a) For each tariff class, the revenue expected to be recovered should lie on or between: (1) an upper bound representing the stand alone cost of serving the customers who belong to that class; and (2) a lower bound representing the avoidable cost of not serving those customers. (b) A tariff, and if it consists of 2 or more charging parameters, each charging parameter for a tariff class: (1) must take into account the long run marginal cost for the service or, in the case of a charging parameter, for the element of the service to which the charging parameter relates; and (2) must be determined having regard to: (i) transaction costs associated with the tariff or each charging parameter; and (ii) whether customers of the relevant tariff class are able or likely to respond to price signals. (c) If, however, as a result of the operation of paragraph (b), the Distribution Network Service Provider may not recover the	(a) Table 2.4 shows that the revenue expected from each tariff class lies between the stand alone cost and the avoidable cost for the tariff class. Section 2.3 contains an explanation of the basis and application of the principle. (b) Section 2.3 contains explanations of how ActewAGL Distribution takes into account long run marginal costs, transactions costs, the scope for customers to respond to price signals and the need to recover revenue in an efficient manner (as required by (c) below). (c) Section 2.3 contains explanations of how the need to recover revenue with minimum distortion to efficient patterns of consumption is taken into account when

	expected revenue, the provider must adjust its tariffs so as to ensure recovery of expected revenue with minimum distortion to efficient patterns of consumption.	setting charging parameters.
6.18.6	(a) This clause applies only to <i>tariff classes</i> related to the provision of <i>standard control services</i>. (b) The expected weighted average revenue to be raised from a <i>tariff class</i> for a particular <i>regulatory year</i> of a <i>regulatory control period</i> must not exceed the corresponding expected weighted average revenue for the preceding <i>regulatory year</i> by more than the permissible percentage. (c) The permissible percentage is the greater of the following: (1) the CPI-X limitation on any increase in the <i>Distribution Network Service Provider's</i> expected weighted average revenue between the two <i>regulatory years</i> plus 2%; Note: The calculation is of the form $(1 + \text{CPI})(1 - X)(1 + 2\%)$ (2) CPI plus 2%. Note: The calculation is of the form $(1 + \text{CPI})(1 + 2\%)$ (d) In deciding whether the permissible percentage has been exceeded in a particular <i>regulatory year</i>, the following are to be disregarded: (1) the recovery of revenue to accommodate a variation to the distribution determination under rule 6.6 or 6.13; (2) the recovery of revenue to accommodate pass through of charges for transmission use of system services to customers. (e) This clause does not, however, limit the extent a tariff for customers with remotely-read interval metering or other similar metering technology may vary according to the time or other circumstances of the customer's usage.	Section 3.2 and Table 3-4 show that tariff classes comply with the side constraints.
6.18.7	(a) A pricing proposal must provide for tariffs designed to pass on to retail customers the designated pricing proposal charges to be incurred by the Distribution Network Service Provider for transmission use of system services. (b) The amount to be passed on to retail customers for a particular <i>regulatory year</i> must not exceed the estimated amount of the designated pricing proposal charges adjusted for over or under recovery in accordance with paragraph (c). (c) The over and under recovery amount must be calculated in a way that: (1) subject to subparagraphs (2) and (3) below, is consistent with the method determined by the AER in the relevant distribution determination for the Distribution Network Service Provider; (2) ensures a Distribution Network Service Provider is able to recover from retail customers no more and no less than the designated pricing proposal charges it incurs; and (3) adjusts for an appropriate cost of capital that is consistent with the allowed rate of return used in the relevant distribution determination for the relevant <i>regulatory year</i>.	Section 3.4 provides an explanation of how TUOS charges are passed on to customers, and ActewAGL Distribution's adjustment for under recovery of TUOS costs in 2012/13. The over and under recovery is shown in Table 3-6. TUOS charges are provided in Table 3.7.

	(d) Notwithstanding anything else in this clause 6.18.7, a Distribution Network Service Provider may not recover charges under this clause to the extent these are: (1) recovered through the Distribution Network Service Provider's annual revenue requirement; (2) recovered under clause 6.18.7A; or (3) recovered from another Distribution Network Service Provider.	
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Table A1.2: Checklist of requirements from the AER final decision

Final decision requirement	Coverage in this document
<i>Compliance with X factors</i> "In accordance with clause 6.12.1(11) of the transitional chapter 6 rules the X factors to apply to ActewAGL are set out in table 17.8 of the final decision" (AER Final Decision, p. 145)	Table 3.1 shows that the X factor applied to the calculation of MAAR for standard control services is -5.41 per cent, consistent with the correction to the AER's final determination. Section 4.2 demonstrates that an X factor of 0.0% has been applied in calculating the revenue cap for alternative control services. This complies with the AER's Final Decision.
<i>Compliance with the standard control services control mechanism.</i> "In accordance with clause 6.12.1(13) of the transitional chapter 6 rules compliance with the maximum allowable average revenue cap for standard control services must be demonstrated by ActewAGL using the formulae outlined in section 4.5 and in accordance with appendix F of this final decision." (AER Final Decision, p. 20)	Section 3.1, shows ActewAGL Distribution's application of the formulae set out in section 4.5 of the AER final decision to derive the revenue cap for standard control services. Section 3.2 and Table 3.3 demonstrate that the revenue from standard control services derived from 2013/14 tariffs complies with this cap. No adjustments under Appendix F have been made. Time-of-use prices have been made the default tariff for new premises and this data will feed into the pricing load profile over time.
<i>Compliance with the alternative control services control mechanism.</i> "In accordance with clause 6.12.1(13) of the transitional chapter 6 rules ActewAGL must demonstrate compliance with the control mechanism for alternative control services by submitting to the AER a schedule of metering charges, in the form of table 13.5 of ActewAGL's revenue proposal, as soon as practicable after prices for each regulatory year are determined." (AER Final Decision, p. 123)	Section 4.2 contains Table 4-2 which is in the same format as Table 13.5 in ActewAGL Distribution's regulatory proposal.
<i>TUOS unders and overs account.</i> "In accordance with clause 6.12.1(19) of the transitional chapter 6 rules ActewAGL must submit, as part of its annual pricing proposal, a record of the amount of revenues recovered from TUOS charges and associated payments in accordance with appendix E of this final decision." (AER Final Decision, p. 20)	The TUOS unders and overs account is shown in section 3.4 (Table 3-6).
<i>Side constraints</i> "In accordance with clause 6.18.2(b)(7) and (8) of the transitional chapter 6 rules ActewAGL must ensure its annual pricing proposals comply with the side constraint formula defined in section 4.5 of this final decision." (AER final decision, p. 20)	Side constraints are considered in section 3.2 and compliance is shown in Table 3-4.